

Conditional Independence Testing beyond i.i.d. Data: A Taxonomy and Residual-Score Framework

Qingzi Huo, Marco Simnacher, Xiaotian Xu, and Sonja Greven
for the Alzheimer’s Disease Neuroimaging Initiative

Working manuscript, 24 August 2026

Abstract

Conditional-independence testing (CIT) with non-i.i.d. data is not one problem: the valid conditioning information, the independent sampling unit, and the object being tested all change across applications. We organise these differences into five cases. Case A concerns state-level hypotheses $X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}$ under three asymptotic regimes: many short clusters, one or a few long sequences, and double asymptotics. Case B adds stable subject information θ_i ; Case C adds observed history H_{it} ; Cases D and E concern conditional independence between entire trajectories, without and with subject information. This taxonomy separates hypotheses that only require an enlarged conditioning set from those that require latent-variable identification or genuinely temporal inference.

We then develop a residual-score framework for Cases A–C. In the theoretically developed many-subject regime, nuisance regressions are cross-fitted by subject, visit-level residual products are aggregated into one score per subject, and inference is based on the $\sqrt{N}$ distribution of independent subject scores. Generalised additive mixed models are one nuisance learner, not a defining component: other regressions or pre-trained embeddings may be used when their prediction errors satisfy the stated product-rate and cross-term conditions. Observed subject information in Case B can be added to the conditioning set; latent or estimated subject information requires separate identification and rate conditions. Case C has two internal regimes: a fixed bounded history with many subjects (C1), which can be included as observed context, and a growing filtration or one/few long series (C2), which requires temporal weak-dependence or martingale theory outside the present cluster theorem.

Simulation evidence reflects these boundaries. The core Case-A null yielded 47 rejections in 1,000 repetitions, but a random-slope stress setting failed the pre-specified calibration gate. New Case-B and Case-C1 simulations distinguish valid observed-context analyses from sparse latent-context estimation and incomplete-history reductions. A supplementary gradient shows how Case-B calibration depends on the accuracy of an independently estimated subject context and provides local-power curves for observed Case B and complete-history C1. In ADNI applications, MRI, plasma biomarkers, amyloid PET, cognition, genetics, and acquisition variables instantiate Case A with many participants and sparse visits. The analyses show how incremental-information conclusions change with the reference set and identify p-tau217 and A β 42/40 as plasma markers retaining information about PET amyloid beyond cognition, demographics, APOE, and acquisition context. A prospective observed-history analysis further identified p-tau217 as the only plasma marker retaining unique information about future ADAS-Cog13. Repeating the subject-fold assignment preserved the plasma-amyloid marker distinction in all 20 assignments and the prospective p-tau217 direction in all 20, although one prospective split crossed the multiplicity-adjusted 0.05 threshold. Cases D and E, and single-series temporal inference, remain future work.

Contents

1	Introduction	4
2	A taxonomy of conditional-independence problems beyond i.i.d. data	5
2.1	Case A: the same null under three asymptotic regimes	5
2.2	Case B: subject information is part of the hypothesis	6
2.3	Case C: observed history under two internal regimes	6
2.4	Cases D and E: trajectory-level hypotheses	7
2.5	Decision rule supplied by the taxonomy	7
3	Residual-score framework for Cases A–C	8
3.1	Many-subject data and the expanded conditioning set	8
3.2	Population conditional means and the GCM target	8
4	Procedure for Case A1 and reducible Cases B and C	9
4.1	Testing target	9
4.2	Population-level nuisance estimation	9
4.3	Cluster-cross-fitted procedure	10
4.4	Interpretation	10
4.5	Several markers and a global multiplier test	10
5	Theoretical framework for Case A1	11
5.1	Notation for cross-fitted prediction errors	11
5.2	High-level assumptions	11
5.3	Null theorem and proof roadmap	12
5.4	Alternative theorem and power	13
6	Score, quasi-score, and exponential-family extensions	14
6.1	General conditionally centred score	14
6.2	Exponential-family outcome	14
6.3	Quasi-score	15
6.4	Pearson residuals and estimated weights	15
6.5	Likelihood and deviance transformations	16
6.6	Distributional score vector	16
7	Sample splitting and cross-fitting	16
7.1	Why splitting must be by subject	16
7.2	Recommended K -fold procedure	16
7.3	What cross-fitting solves and what it does not	16
8	Relevant theory of the original GCM	17
8.1	Original univariate statistic	17
8.2	Theorem 6: null validity	17
8.3	Theorem 8: alternatives and power	18
8.4	Theorem 9: multivariate and high-dimensional GCM	18
8.5	Theorem 11: kernel ridge regression	18
9	Case B: observed, estimated, and latent subject context	19
9.1	Observed context reduces to Case A	19
9.2	A different scientific target	19
9.3	Latent-effect identification	19
9.4	Finite visits and irreducible prediction error	19

9.5	BLUP shrinkage can create residual dependence	19
9.6	Why sample splitting is insufficient	20
9.7	Dense-visit regime	20
9.8	Recommended interpretation	20
10	Case C: bounded and growing observed history	20
10.1	C1: many subjects with bounded histories	20
10.2	C2: growing history in one or a few long sequences	21
10.3	High-dimensional measurements and learned representations	21
11	Simulation study	21
11.1	Objectives and design	21
11.2	Why the independent unit matters	22
11.3	Rejection rates under the null	22
11.4	Observed context, estimated context, and bounded history	23
12	ADNI applications in the many-subject, sparse-visit regime	24
12.1	Scientific question and data construction	24
12.2	Analysis	25
12.3	Results	26
12.4	What the existing analyses distinguish	26
12.5	Plasma information about PET amyloid beyond cognition and genetics	28
12.6	Prospective plasma information about future cognition given observed history	29
12.7	Additional ADNI conditional pathology-information targets	30
13	Discussion	31
14	Reproducibility, data availability, and governance	33
15	Limitations and next directions	33
16	Scope and evidence summary	34

1 Introduction

Many biomedical, behavioural, and sensor studies ask an incremental-information question: does a candidate measurement tell us something about an outcome after the information already available from clinical, behavioural, demographic, or technical variables has been taken into account? Conditional-independence testing gives this question a direct statistical form. The generalised covariance measure (GCM) estimates two conditional means and tests whether the product of their residuals has expectation zero [1]. Its flexibility comes from allowing a broad class of regression methods, but its standard theory is formulated for independent observations.

The phrase “non-i.i.d. CIT” hides several distinct problems. Repeated measurements from many subjects create independent clusters; one long time series instead has no replication across subjects; stable subject information may be observed, estimated, or latent; and a null about current measurements is not the same as a null about complete trajectories. These distinctions determine what must enter the conditioning set, what is asymptotically independent, and which central limit theorem can justify a reference distribution. Treating all of them as a generic mixed-model correction changes the hypothesis without resolving the sampling problem.

We begin with a five-case taxonomy organised by the object of inference and the available information. It then develops a common residual-score construction for the cases that can be reduced to conditional-mean estimation plus an appropriate score-level central limit theorem. For many independent subjects, visit-level residual products are aggregated into one score per subject. Nuisance models are cross-fitted by subject, predictions for held-out subjects exclude subject-specific BLUPs in a population analysis, and inference uses the empirical variance of complete subject scores. A Gaussian multiplier extends the construction to several candidate markers.

The three contributions are:

1. **A problem map for non-i.i.d. CIT.** We distinguish five nulls (Cases A–E), three asymptotic regimes inside Case A, and observed versus latent subject information inside Case B. The map makes the required domain knowledge and the unsolved identification problems explicit.
2. **A residual-score framework with an explicit scope.** We develop the cluster construction and high-level rate conditions for Case A1. Case B with observed subject information and Case C1 with complete bounded history can be reduced to Case A1 by enlarging the conditioning set. Estimated or latent subject information, growing-history Case C2, and double asymptotics require additional assumptions and are presented separately. The nuisance learner may be a GAMM, random forest, kernel method, neural network, or a pre-trained embedding, provided the relevant prediction and leakage conditions hold.
3. **Evidence across the map.** Existing simulations evaluate Case A and expose a random-slope limitation and a failed sparse-BLUP construction. New simulations isolate the roles of observed or estimated subject context and complete or incomplete history, quantify the proxy-accuracy gradient for generated subject context, and supply local-power curves. ADNI analyses instantiate the many-subject, sparse-visit regime and evaluate whether imaging and plasma measurements add information beyond explicitly named clinical, historical, and genetic reference sets. A prospective application links this question to future cognitive status, and repeated subject-fold analyses assess dependence on one cross-fitting split.

Two boundaries are maintained throughout. First, the scalar residual score targets an expected conditional covariance: conditional independence implies a zero target, but the converse need not hold. Second, incremental information is not a causal effect. Rich transformations, multivariate scores, or embedding-based CITs can broaden the detected alternative class, but they do not remove the need to match the null, conditioning set, and sampling regime.

2 A taxonomy of conditional-independence problems beyond i.i.d. data

Let i index subjects or clusters and t index occasions. Write $\bar{X}_{it} = (X_{i1}, \dots, X_{it})$, with analogous notation for $\bar{Y}_{it}$ and $\bar{Z}_{it}$. Let H_{it} denote pre-specified observed history available before the tested response, and let θ_i denote stable subject information. The five cases differ in the random object being tested and in whether the additional context is observed.

For i.i.d. observations, the literature can be grouped by what is learned under the null. Kernel and operator methods encode conditional association in reproducing-kernel Hilbert spaces [2, 3]; residual and prediction methods compare conditional means or risks, including GCM and its weighted extension [1, 4]; and model-X or generative methods approximate a conditional distribution [5]. The GCM deliberately trades omnibus power for transparent regression-rate requirements. Non-i.i.d. extensions must add a second layer: a valid approximation for the dependent score process. This is not a cosmetic variance correction. For example, an i.i.d. bootstrap for kernel independence can fail for random processes even when the dependence statistic itself is unchanged [6]. Shi’s GTCM uses stability and weak temporal dependence for a history-conditioned predictive null [8]; Wieck-Sosa, Haddad, and Ramdas develop time-varying regression, time-varying covariance estimation, and Gaussian approximation for a single non-stationary nonlinear series [9]. DNCIT addresses a different axis, replacing complex images by learned representations before applying a downstream CIT [10]. Table 2 locates these contributions by sampling regime and information structure.

The comparison in Table 1 separates two questions that are often combined. The first is how conditional dependence is represented: kernels, residual scores, conditional generators, or learned features. The second is how the reference distribution accounts for dependence across observations. A method can address the first question without answering the second. The present framework changes the second layer for A1 and leaves the choice of nuisance learner open.

2.1 Case A: the same null under three asymptotic regimes

Case A uses the same state-level null but not the same reference distribution.

A1: $N \rightarrow \infty$, bounded or controlled T_i .

Independent subjects are the asymptotic units. Arbitrary within-subject dependence can be retained inside a subject score, subject-level cross-fitting prevents leakage, and a cluster central limit theorem supplies $\sqrt{N}$ inference. ADNI’s many participants and sparse visits belong to this regime.

A2: fixed or small N , $T \rightarrow \infty$.

The asymptotic information comes from time. A valid test requires temporal weak dependence, mixing or martingale conditions, a temporal variance estimate, and a fitting scheme whose dependence on the tested innovations is controlled. The Generalised Temporal Covariance Measure presented by Shi is an example for a predictive null involving Y_{t+1} , X_t , and observed history [8].

A3: $N \rightarrow \infty$ and $T_i \rightarrow \infty$.

Both dimensions contribute. One must control the largest cluster influence, specify how N and T_i grow, and combine cluster and temporal approximation errors. Neither the A1 theorem nor an A2 temporal theorem automatically proves this regime.

Table 1: Representative CIT and dependence-testing approaches organised by statistical object and sampling regime. The table is selective: its purpose is to show which component of the non-i.i.d. problem each method addresses.

Approach	Statistical object	Sampling regime	Main calibration requirement	Relation to A–E
Kernel CI and KCI [2, 3]	RKHS conditional covariance operator	independent triples	kernel/operator regularity and an i.i.d. null approximation	baseline CIT; no cluster or temporal layer
GCM and WGCM [1, 4]	products of conditional-mean residuals, with optional weights	independent triples	nuisance prediction rates and score CLT	algebraic basis for the A1 score
Generative CIT [5]	draws from an estimated conditional null	independent, potentially high dimensional	conditional-distribution approximation	alternative nuisance engine; non-i.i.d. calibration still required
Random-process HSIC [6]	instantaneous unconditional process independence	one dependent process	process assumptions and dependent-data bootstrap	shows why i.i.d. resampling fails
GTCM [8]	predictive residual score given growing observed history	one long sequence	regression stability and weak temporal dependence	A2/C2
Time-varying single-series CIT [9]	nonlinear CI score	one nonstationary series	time-varying nuisance and covariance approximation	A2/C2
DNCIT [10]	learned representation followed by a downstream CIT	high-dimensional images	representation training must not create conditional dependence	orthogonal to sampling; composable with A–C
Subject-score framework (this article)	within-subject mean of cross-fitted residual products	many independent subjects, bounded visits	subject splitting, product rates, and cluster CLT	A1; observed B and C1 reduce to A1

Table 2: Five cases of non-i.i.d. conditional-independence testing. “Current status” refers to the scope of this article, not to the absence of other specialised methods.

Case	Null hypothesis	Additional information	Natural independent unit	Current status
A	$X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}$	observed current context	subject or time block	A1 developed; A2/A3 separated
B	$X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}, \theta_i$	stable observed or latent subject context	subject	observed subcase reducible; latent subcase conditional
C	$X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}, H_{it}$	observed history	subject or temporal block	C1 reducible; C2 temporal/heuristic here
D	$\tilde{X}_{it} \perp\!\!\!\perp \tilde{Y}_{it} \mid Z_{it}$	complete observed paths	trajectory	future work
E	$\tilde{X}_{it} \perp\!\!\!\perp \tilde{Y}_{it} \mid Z_{it}, \theta_i$	paths plus stable subject context	trajectory	future work

2.2 Case B: subject information is part of the hypothesis

If θ_i is observed before the tested outcomes, it can be appended to Z_{it} and the problem reduces to the appropriate Case-A regime. This includes fixed baseline attributes, validated genetic variables, and externally measured subject descriptors. If θ_i is estimated, it is a generated conditioning variable. Validity then requires both scientific identification and sufficiently small estimation error; the residual-score product makes the product of the two nuisance errors particularly important. If θ_i is latent and each subject has only a few observations, it is generally not consistently recoverable without a structural model. A BLUP is therefore not interchangeable with an observed covariate.

2.3 Case C: observed history under two internal regimes

Case C is one top-level hypothesis but contains two inferential regimes. In **C1**, H_{it} is finite-dimensional, observed, fixed before the tested response, and uniformly bounded as the number

of subjects grows. It can then be appended to Z_{it} , so the many-subject problem reduces to A1 even if raw visits are serially dependent. In **C2**, H_{it} grows with t or denotes the filtration generated by an increasingly long past. The residual scores are then temporally dependent and an A2 temporal theorem (or an A3 double-asymptotic theorem) is required. Shi’s GTCM work is closer to C2 because its predictive null conditions on $\bar{Y}_t$ and $\bar{Z}_t$, rather than on a fixed one-lag vector [8]. The current prospective ADNI analysis is C1. Omitting a relevant lag changes the null being tested rather than only reducing efficiency.

2.4 Cases D and E: trajectory-level hypotheses

Cases D and E concern independence between stochastic paths, not same-time residual moments. They are stronger and typically high-dimensional hypotheses. A scalar score at each visit is not an omnibus test for them. Embeddings, kernel mean representations, or functional scores may reduce the trajectories, but the transformation must not use the tested outcomes in a way that creates dependence. DNCIT provides a useful template for composing a learned embedding with a downstream CIT under explicit conditions on how the embedding was trained [10]; the present article does not yet supply the trajectory-level calibration needed for Cases D and E.

2.5 Decision rule supplied by the taxonomy

The practical order is: define the null; identify whether the target is state-level or trajectory-level; list the information that must be conditioned on; determine whether that information is observed, estimated, or latent; and only then select the independent unit and the nuisance learner. This order prevents a convenient regression model from silently redefining the scientific question (Figure 1).

Five top-level non-i.i.d. conditional-independence targets

Choose the target and information set before the nuisance learner or reference distribution.

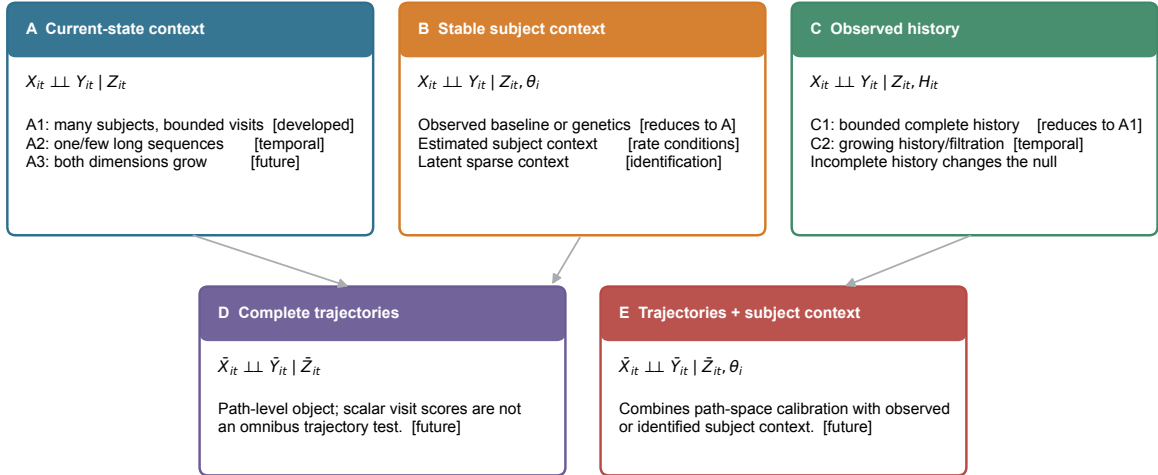


Figure 1: Five top-level non-i.i.d. conditional-independence targets. A1–A3 and C1–C2 are inferential regimes within Cases A and C, respectively, not additional top-level cases. Blue and green subcases are developed or reducible to the subject-score construction under the stated information conditions; temporal, latent, and trajectory-level branches require additional identification or reference-distribution theory.

3 Residual-score framework for Cases A–C

3.1 Many-subject data and the expanded conditioning set

Let $i = 1, \dots, N$ index independent subjects and let $t = 1, \dots, m_i$ index repeated visits within subject i . We observe

$$\mathcal{O}_i = \{(X_{it}, Y_{it}, Z_{it}) : t = 1, \dots, m_i\}.$$

The vectors $\mathcal{O}_1, \dots, \mathcal{O}_N$ are assumed independent, while arbitrary serial or exchangeable dependence is allowed within each $\mathcal{O}_i$ unless a more specific assumption is imposed.

The primary developed regime is A1: N independent subjects and a bounded or suitably controlled number of observations per subject. The notation Z_{it} below denotes the *complete observed conditioning vector* required by the scientific question. It can be replaced by (Z_{it}, θ_i) in observed-Case B or by (Z_{it}, H_{it}) in multi-subject Case C1. This reduction is valid only when the added information is observed without outcome leakage and has manageable dimension.

The baseline hypothesis is the visit-level, population-level conditional independence statement

$$H_0^{\text{pop}} : X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}, \quad t = 1, \dots, m_i. \quad (1)$$

Here “population-level” means that subject-specific latent effects are integrated out; they are not conditioned upon and are not removed through subject-specific BLUPs.

For a prospective question, the aligned target is

$$H_{0,\Delta}^{\text{pop}} : X_{it} \perp\!\!\!\perp Y_{i,t+\Delta} \mid Z_{it}, \quad (2)$$

where Δ is a pre-specified forecast horizon and Z_{it} contains only information available at the origin time. The same subject-score construction applies after replacing Y_{it} by its aligned future outcome. This formulation turns biomarker or sensor evaluation into a question about information added beyond an explicit reference set.

Time, calendar day, baseline covariates, treatment, environmental variables, and relevant history should be included in Z_{it} whenever they are scientifically required. This is important because the interpretation of (1) is always relative to the chosen conditioning information.

The present theorem is aimed at many independent subjects with a moderate number of repeated observations per subject. Very long, high-frequency sequences require an A2 or A3 method that combines explicit temporal dependence control with cluster-level inference; pre-specified daily or session summaries may be scientifically useful, but do not by themselves prove the required conditional-independence statement.

3.2 Population conditional means and the GCM target

Define

$$f_0(z) = \mathbb{E}(X_{it} \mid Z_{it} = z), \quad g_0(z) = \mathbb{E}(Y_{it} \mid Z_{it} = z),$$

and the marginal residuals

$$\varepsilon_{it} = X_{it} - f_0(Z_{it}), \quad \xi_{it} = Y_{it} - g_0(Z_{it}).$$

Let $w_{it} \geq 0$ denote pre-specified within-subject weights satisfying

$$\sum_{t=1}^{m_i} w_{it} = 1.$$

The equal-subject choice is $w_{it} = m_i^{-1}$. Define the oracle subject score

$$S_i = \sum_{t=1}^{m_i} w_{it} \varepsilon_{it} \xi_{it}. \quad (3)$$

The corresponding population parameter is

$$\rho = \mathbb{E}(S_i) = \mathbb{E} \left[\sum_{t=1}^{m_i} w_{it} \{X_{it} - f_0(Z_{it})\} \{Y_{it} - g_0(Z_{it})\} \right]. \quad (4)$$

Under (1), each same-visit conditional covariance is zero and hence $\rho = 0$.

Remark 3.1 (What is and is not tested). Conditional independence implies $\rho = 0$, but the converse does not hold in general. The scalar GCM is therefore a valid conditional-independence test with power against alternatives for which the weighted expected conditional covariance is non-zero. Richer transformations or multivariate scores are needed to detect conditional dependence that has zero residual covariance.

Remark 3.2 (Equal-subject versus equal-visit targets). The choice $w_{it} = m_i^{-1}$ gives each subject equal total weight. An equal-visit target can instead be constructed by weighting subjects proportionally to m_i . If cluster size is informative, the two targets differ scientifically and must not be treated as interchangeable.

4 Procedure for Case A1 and reducible Cases B and C

4.1 Testing target

After constructing the scientifically complete observed context, write it as Z_{it} and test $H_0 : X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}$. The same algorithm is therefore used for A1, for Case B with observed θ_i , and for multi-subject Case C1 with complete finite H_{it} . What changes is the meaning and content of Z_{it} , not the score formula.

4.2 Population-level nuisance estimation

Estimate f_0 and g_0 with flexible regression procedures that account for repeated measurements during fitting. GAMMs are one option, but the target of prediction must remain $\mathbb{E}(X_{it} \mid Z_{it})$ and $\mathbb{E}(Y_{it} \mid Z_{it})$.

For a Gaussian identity-link GAMM [11], population-level predictions normally exclude the subject-specific BLUP. For a nonlinear link, simply omitting the random effect on the linear predictor scale does not generally equal the marginal mean. The marginal prediction should then integrate over the fitted random-effect distribution.

4.3 Cluster-cross-fitted procedure

Procedure 1: Cluster-robust marginal GCM

1. Split the N subjects into K folds $I_1, \dots, I_K$. All visits from one subject must remain in the same fold.

2. For fold k , use subjects in I_k^c to fit

$$\hat{f}^{(-k)}(z) \approx \mathbb{E}(X \mid Z = z), \quad \hat{g}^{(-k)}(z) \approx \mathbb{E}(Y \mid Z = z).$$

If a supervised neural-network embedding is part of X , train the network, choose the embedding layer, and perform all dimension reduction using training subjects only.

3. For each held-out subject $i \in I_k$, compute population-level residuals

$$\hat{\varepsilon}_{it} = X_{it} - \hat{f}^{(-k)}(Z_{it}), \quad \hat{\xi}_{it} = Y_{it} - \hat{g}^{(-k)}(Z_{it}).$$

Do not include held-out subject-specific BLUPs in a marginal analysis.

4. Construct the held-out subject score

$$\hat{S}_i = \sum_{t=1}^{m_i} w_{it} \hat{\varepsilon}_{it} \hat{\xi}_{it}.$$

5. Pool the cross-fitted scores and compute

$$\bar{S}_N = \frac{1}{N} \sum_{i=1}^N \hat{S}_i, \quad \hat{\sigma}_N^2 = \frac{1}{N} \sum_{i=1}^N (\hat{S}_i - \bar{S}_N)^2.$$

6. Form

$$T_N = \frac{\sqrt{N} \bar{S}_N}{\hat{\sigma}_N}. \tag{5}$$

Reject the two-sided null at level α when $|T_N| > z_{1-\alpha/2}$.

4.4 Interpretation

The first-stage regressions remove the population-level dependence of X and Y on Z . The subject score aggregates all visit-level residual products within one independent cluster. The denominator estimates the variance of the complete subject score, and therefore includes all within-subject covariance terms automatically.

This differs fundamentally from treating all $M = \sum_i m_i$ visits as independent and using $\sqrt{M}$ scaling. The effective asymptotic unit is the subject, so the basic scaling is $\sqrt{N}$.

4.5 Several markers and a global multiplier test

For markers $X^{(1)}, \dots, X^{(p)}$, let $\hat{S}_{ij}$ be the cross-fitted subject score for marker j , with mean $\bar{S}_j$ and empirical standard deviation $\hat{\sigma}_j$. The marker statistic is

$$T_j = \frac{\sqrt{N} \bar{S}_j}{\hat{\sigma}_j}, \quad j = 1, \dots, p,$$

and the global statistic is $M = \max_j |T_j|$. For multiplier draw b , generate independent standard normal variables,

$$e_i^{(b)} \sim N(0, 1), \quad i = 1, \dots, N,$$

and calculate

$$T_j^{*(b)} = \frac{1}{\sqrt{N}\hat{\sigma}_j} \sum_{i=1}^N e_i^{(b)}(\hat{S}_{ij} - \bar{S}_j), \quad M^{*(b)} = \max_j |T_j^{*(b)}|.$$

With B multiplier draws, the empirical global p -value is

$$\hat{p}_{\text{global}} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{M^{*(b)} \geq M\}}{B + 1}.$$

The same multiplier is applied to all markers within a subject, thereby preserving their estimated dependence. In the application, the global test is primary and normal-reference marker tests are adjusted by Holm's method.

5 Theoretical framework for Case A1

5.1 Notation for cross-fitted prediction errors

For subject i in fold $k(i)$, define

$$\Delta_{it}^f = \hat{f}^{(-k(i))}(Z_{it}) - f_0(Z_{it}), \quad \Delta_{it}^g = \hat{g}^{(-k(i))}(Z_{it}) - g_0(Z_{it}).$$

Then

$$\hat{\varepsilon}_{it}\hat{\xi}_{it} = \varepsilon_{it}\xi_{it} - \varepsilon_{it}\Delta_{it}^g - \xi_{it}\Delta_{it}^f + \Delta_{it}^f\Delta_{it}^g. \quad (6)$$

At subject level,

$$\hat{S}_i = S_i - C_{g,i} - C_{f,i} + B_i, \quad (7)$$

where

$$C_{g,i} = \sum_t w_{it}\varepsilon_{it}\Delta_{it}^g, \quad C_{f,i} = \sum_t w_{it}\xi_{it}\Delta_{it}^f, \quad B_i = \sum_t w_{it}\Delta_{it}^f\Delta_{it}^g.$$

5.2 High-level assumptions

Assumption 5.1 (Independent clusters). The subject vectors $\mathcal{O}_1, \dots, \mathcal{O}_N$ are independent. They may be non-identically distributed in a triangular array, provided the following moment and variance conditions hold uniformly.

Assumption 5.2 (Weights and cluster size). The weights are nonnegative and satisfy $\sum_t w_{it} = 1$. Either $\max_i m_i \leq m_{\max} < \infty$, or growing clusters satisfy a Lindeberg condition that prevents a single subject from dominating the total score.

Assumption 5.3 (Oracle centering). For every visit,

$$\mathbb{E}(\varepsilon_{it} \mid Z_{it}) = 0, \quad \mathbb{E}(\xi_{it} \mid Z_{it}) = 0.$$

Under the null, $X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}$ and therefore $\mathbb{E}(\varepsilon_{it}\xi_{it} \mid Z_{it}) = 0$.

Assumption 5.4 (Oracle cluster CLT). For some $\delta > 0$,

$$\sup_i \mathbb{E}|S_i - \mathbb{E}S_i|^{2+\delta} < \infty,$$

and

$$\sigma_N^2 = \frac{1}{N} \sum_{i=1}^N \text{Var}(S_i) \longrightarrow \sigma^2 \in (0, \infty).$$

The corresponding Lindeberg condition holds. For identically distributed clusters this reduces to $0 < \text{Var}(S_i) < \infty$ plus a suitable $2 + \delta$ moment.

Assumption 5.5 (Cluster cross-fitting). For every held-out cluster i , the functions $\hat{f}^{(-k(i))}$ and $\hat{g}^{(-k(i))}$ are estimated without using any observation from subject i . Conditional on the corresponding fold-specific training data, the prediction errors at a held-out visit are functions of the held-out Z_{it} , not of held-out X_{it} or Y_{it} .

Assumption 5.6 (Product-rate condition). Define the cluster-weighted mean squared prediction errors

$$A_f = \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^{m_i} w_{it} (\Delta_{it}^f)^2, \quad A_g = \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^{m_i} w_{it} (\Delta_{it}^g)^2.$$

Assume

$$A_f A_g = o_{\mathbb{P}}(N^{-1}). \quad (8)$$

Equivalently, if $r_f = A_f^{1/2}$ and $r_g = A_g^{1/2}$ are empirical RMSEs, then $\sqrt{N} r_f r_g = o_{\mathbb{P}}(1)$.

Assumption 5.7 (Cluster-weighted cross-term control). Let $n_k = |I_k|$ and let $\mathcal{T}_k$ denote the observations from subjects outside fold I_k , together with the fold assignment. For fixed K , assume

$$B_{g,k}^{\text{cl}} = \frac{1}{n_k} \sum_{i \in I_k} \mathbb{E} \left[\left(\sum_t w_{it} \varepsilon_{it} \Delta_{it}^g \right)^2 \middle| \mathcal{T}_k \right] \quad \max_{k \leq K} B_{g,k}^{\text{cl}} = o_{\mathbb{P}}(1), \quad (9)$$

$$B_{f,k}^{\text{cl}} = \frac{1}{n_k} \sum_{i \in I_k} \mathbb{E} \left[\left(\sum_t w_{it} \xi_{it} \Delta_{it}^f \right)^2 \middle| \mathcal{T}_k \right] \quad \max_{k \leq K} B_{f,k}^{\text{cl}} = o_{\mathbb{P}}(1). \quad (10)$$

Remark 5.1 (A sufficient covariance-operator condition). Let $\Delta_i^g = (\Delta_{i1}^g, \dots, \Delta_{im_i}^g)^\top$, $W_i = \text{diag}(w_{i1}, \dots, w_{im_i})$, and let Ω_i^X denote the conditional or unconditional covariance matrix of $(\varepsilon_{i1}, \dots, \varepsilon_{im_i})^\top$ under the conditioning used in the proof. Then a sufficient version of (9), applied uniformly over the fixed folds, is

$$\max_{k \leq K} \frac{1}{n_k} \sum_{i \in I_k} (\Delta_i^g)^\top W_i \Omega_i^X W_i \Delta_i^g = o_{\mathbb{P}}(1),$$

with the analogous condition for f . If the operator norms of Ω_i^X and Ω_i^Y are uniformly bounded, ordinary cluster-weighted L_2 consistency is sufficient for these two conditions.

Assumption 5.8 (Variance consistency). The estimated subject scores satisfy

$$\frac{1}{N} \sum_{i=1}^N (\hat{S}_i - S_i)^2 = o_{\mathbb{P}}(1),$$

and a law of large numbers holds for S_i^2 . Consequently,

$$\hat{\sigma}_N^2 - \frac{1}{N} \sum_i (S_i - \bar{S}_N^0)^2 = o_{\mathbb{P}}(1),$$

where $\bar{S}_N^0 = N^{-1} \sum_i S_i$.

5.3 Null theorem and proof roadmap

Theorem 5.2 (Proposed null validity of cluster GCM). *Under Assumptions 4.1–4.8 and H_0^{pop} , the statistic in (5) satisfies*

$$T_N \xrightarrow{d} N(0, 1).$$

If all assumptions hold uniformly over a class of null distributions, then the convergence is uniform and the resulting test has uniformly asymptotic level α over that class.

Proof roadmap. Using (7),

$$\sqrt{N} \bar{S}_N = \frac{1}{\sqrt{N}} \sum_{i=1}^N S_i - \frac{1}{\sqrt{N}} \sum_{i=1}^N C_{g,i} - \frac{1}{\sqrt{N}} \sum_{i=1}^N C_{f,i} + \frac{1}{\sqrt{N}} \sum_{i=1}^N B_i. \quad (11)$$

The proof consists of four steps.

Step 1: Oracle cluster CLT. Under the null, $\mathbb{E}S_i = 0$. Assumption 5.4 gives

$$\frac{N^{-1/2} \sum_i S_i}{\sigma_N} \xrightarrow{d} N(0, 1).$$

Within-subject dependence is entirely contained in S_i and does not violate the CLT because subjects are independent.

Step 2: Product-bias term. By Cauchy-Schwarz and $\sum_t w_{it} = 1$,

$$\begin{aligned} \left| \frac{1}{\sqrt{N}} \sum_i B_i \right| &\leq \sqrt{N} \left\{ \frac{1}{N} \sum_{i,t} w_{it} (\Delta_{it}^f)^2 \right\}^{1/2} \left\{ \frac{1}{N} \sum_{i,t} w_{it} (\Delta_{it}^g)^2 \right\}^{1/2} \\ &= \sqrt{N} (A_f A_g)^{1/2} = o_{\mathbb{P}}(1). \end{aligned}$$

Step 3: Two first-order cross terms. For each fold k , cluster cross-fitting ensures that, conditional on $\mathcal{T}_k$, held-out nuisance functions do not depend on held-out responses. Since $\mathbb{E}(\varepsilon_{it} \mid Z_{it}) = 0$ and $\mathbb{E}(\xi_{it} \mid Z_{it}) = 0$, each cross term is centred after conditioning on the appropriate fold-specific training information. Subjects within I_k are conditionally independent given $\mathcal{T}_k$, and Assumption 5.7 gives

$$\frac{1}{\sqrt{n_k}} \sum_{i \in I_k} C_{g,i} = o_{\mathbb{P}}(1), \quad \frac{1}{\sqrt{n_k}} \sum_{i \in I_k} C_{f,i} = o_{\mathbb{P}}(1)$$

by conditional Chebyshev's inequality. Summing the fixed number of folds with weights $\sqrt{n_k/N}$ yields the two $N^{-1/2}$ results required in (11). This foldwise argument avoids conditioning jointly on overlapping training samples.

Step 4: Studentisation. Assumption 5.8 implies $\hat{\sigma}_N^2 \rightarrow_{\mathbb{P}} \sigma^2$. Slutsky's theorem completes the proof. $\square$

5.4 Alternative theorem and power

Under a general alternative, define

$$\rho_N = \frac{1}{N} \sum_{i=1}^N \mathbb{E}(S_i), \quad \sigma_{N,A}^2 = \frac{1}{N} \sum_{i=1}^N \text{Var}(S_i).$$

For identically distributed clusters, $\rho_N = \rho$ and $\sigma_{N,A}^2 = \text{Var}(S_i)$.

Theorem 5.3 (Proposed asymptotic linearity under alternatives). *Assume cluster cross-fitting and the analogues of Assumptions 4.4–4.8 for the centred oracle scores $S_i - \mathbb{E}S_i$. Then*

$$\frac{\sqrt{N}(\bar{S}_N - \rho_N)}{\hat{\sigma}_N} \xrightarrow{d} N(0, 1).$$

Consequently, fixed alternatives with $\rho \neq 0$ are detected consistently. Under local alternatives $\rho_N = c/\sqrt{N}$, the limiting distribution is normal with mean c/σ and variance one.

Remark 5.4 (Why cross-fitting is more important under alternatives). Without splitting, the regression estimate of g_0 is a function of the same Y values used in the test. Under the null, conditional independence may provide a same-sample centering argument. Under an alternative, $\mathbb{E}(\varepsilon \mid Y, Z)$ need not vanish, so the first-order cross term is no longer automatically centred. Cluster cross-fitting restores centering through $\mathbb{E}(\varepsilon \mid Z) = 0$ and $\mathbb{E}(\xi \mid Z) = 0$, properties that hold under both the null and the alternative.

6 Score, quasi-score, and exponential-family extensions

6.1 General conditionally centred score

Let $q_0(Y, Z)$ be any square-integrable transformation satisfying

$$\mathbb{E}\{q_0(Y, Z) \mid Z\} = 0. \quad (12)$$

Define the oracle score-GCM contribution

$$R_{it}^q = \{X_{it} - f_0(Z_{it})\}q_0(Y_{it}, Z_{it}),$$

and the subject score

$$S_i^q = \sum_t w_{it} R_{it}^q.$$

Under $X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}$,

$$\mathbb{E}(R_{it}^q \mid Z_{it}) = \mathbb{E}(\varepsilon_{it} \mid Z_{it})\mathbb{E}\{q_0(Y_{it}, Z_{it}) \mid Z_{it}\} = 0.$$

Thus a conditionally centred score may replace the ordinary Y residual.

Let $\hat{q}_{it}$ be a cross-fitted estimate and define

$$\Delta_{it}^q = \hat{q}_{it} - q_0(Y_{it}, Z_{it}).$$

The estimated product has the exact expansion

$$\{X_{it} - \hat{f}(Z_{it})\}\hat{q}_{it} = \varepsilon_{it}q_{0,it} - \Delta_{it}^f q_{0,it} + \varepsilon_{it}\Delta_{it}^q - \Delta_{it}^f \Delta_{it}^q. \quad (13)$$

Define

$$A_q = \frac{1}{N} \sum_{i,t} w_{it} (\Delta_{it}^q)^2.$$

The direct analogue of the GCM product-rate condition is

$$A_f A_q = o_{\mathbb{P}}(N^{-1}). \quad (14)$$

The corresponding cluster cross-term conditions replace ξ by q_0 and Δ^g by Δ^q in (9)–(10).

6.2 Exponential-family outcome

Suppose the conditional density of Y is

$$p(y \mid Z) = \exp \left[\frac{y\theta(Z) - b\{\theta(Z)\}}{a(\phi)} + c(y, \phi) \right].$$

Then

$$\mu_0(Z) = \mathbb{E}(Y \mid Z) = b'\{\theta(Z)\}, \quad \text{Var}(Y \mid Z) = a(\phi)b''\{\theta(Z)\}.$$

The score for the natural parameter is

$$q_0^{\text{can}}(Y, Z) = \frac{Y - \mu_0(Z)}{a(\phi)}. \quad (15)$$

When $a(\phi)$ is known or is a global bounded scale, it can be omitted without changing the studentised test. With

$$\hat{q}^{\text{can}}(Y, Z) = Y - \hat{\mu}(Z),$$

we have

$$\Delta^q(Z) = \mu_0(Z) - \hat{\mu}(Z),$$

which is a function of Z and training data only. Therefore

$$A_q = \frac{1}{N} \sum_{i,t} w_{it} \{\hat{\mu}(Z_{it}) - \mu_0(Z_{it})\}^2,$$

and the required condition is exactly

$$A_f A_\mu = o_{\mathbb{P}}(N^{-1}). \quad (16)$$

This is the cleanest exponential-family extension: the null and alternative proof structures are the same as for the ordinary GCM.

Remark 6.1 (Rates). If $r_f = A_f^{1/2} = O_{\mathbb{P}}(N^{-\alpha})$ and $r_\mu = A_\mu^{1/2} = O_{\mathbb{P}}(N^{-\beta})$, then (16) requires $\alpha + \beta > 1/2$. If the outcome mean is estimated by a correctly specified finite-dimensional GLM, $r_\mu = O_{\mathbb{P}}(N^{-1/2})$, so consistency of the X regression may be sufficient. If both sides are nonparametric with comparable difficulty, each RMSE typically needs to be $o_{\mathbb{P}}(N^{-1/4})$.

6.3 Quasi-score

A general quasi-score can be written as

$$q_0^Q(Y, Z) = w_0(Z) \{Y - \mu_0(Z)\}, \quad (17)$$

where $w_0(Z)$ is a bounded, pre-specified, or externally estimated weight. Since $\mathbb{E}(q_0^Q \mid Z) = 0$, the null proof remains valid. If w_0 is fixed and bounded, then

$$A_q \lesssim A_\mu$$

and the same product-rate condition applies.

The classical quasi-likelihood choice is

$$w_0(Z) = \frac{\partial \mu_0(Z) / \partial \eta}{\text{Var}(Y \mid Z)},$$

but full distributional correctness is not required for centering if the mean model is correct and the weight is a function of Z only.

6.4 Pearson residuals and estimated weights

The Pearson residual is

$$q_0^P(Y, Z) = \frac{Y - \mu_0(Z)}{\sigma_0(Z)}.$$

It is conditionally centred when $\sigma_0(Z) > 0$. If σ_0 is known or treated as a fixed bounded weight, the theory is the same as the quasi-score theory.

If σ_0 is estimated,

$$\hat{q}^P - q_0^P = \frac{\mu_0 - \hat{\mu}}{\hat{\sigma}} + (Y - \mu_0) \left(\frac{1}{\hat{\sigma}} - \frac{1}{\sigma_0} \right).$$

The second term depends on Y . Under the null, conditional independence can still help centre the corresponding cross term. Under a general alternative, however, independent sample splitting alone does not make this term mean zero because $\mathbb{E}\{\varepsilon(Y - \mu_0) \mid Z\}$ is precisely a conditional covariance. A separate rate, Taylor expansion, or a score orthogonalised with respect to the variance nuisance is then required.

6.5 Likelihood and deviance transformations

A raw log-likelihood contribution $\ell(Y; \theta(Z))$ and a deviance residual are generally not conditionally mean zero. They cannot be inserted into the GCM without centering. A valid transformation is

$$q_0^\ell(Y, Z) = \ell\{Y; \theta_0(Z)\} - \mathbb{E}[\ell\{Y; \theta_0(Z)\} \mid Z].$$

This introduces additional nuisance functions and typically creates Y -dependent estimation errors. It is therefore less attractive than the canonical or fixed-weight quasi-score for a first theoretical development.

6.6 Distributional score vector

To investigate whether X is related to different aspects of the conditional distribution of Y , one may use a vector of centred scores, for example

$$q_{1,0}(Y, Z) = Y - \mu_0(Z), \quad q_{2,0}(Y, Z) = \{Y - \mu_0(Z)\}^2 - \text{Var}(Y \mid Z).$$

The resulting multivariate GCM can test association with the conditional mean and dispersion simultaneously. Inference then follows from a fixed-dimensional Wald argument or a high-dimensional maximum-statistic argument.

7 Sample splitting and cross-fitting

7.1 Why splitting must be by subject

All visits from one subject must be kept in the same fold. Visit-level random splitting leaks subject-specific information between the training and test sets, invalidates the interpretation of an out-of-sample score, and can make the nuisance estimators dependent on the held-out cluster score.

Cluster splitting also avoids a subtle strengthening of the null. If $\hat{g}$ is estimated using all Y values from subject i , controlling $\varepsilon_{it}\{g(Z_{it}) - \hat{g}(Z_{it})\}$ may require information about $\mathbb{E}(\varepsilon_{it} \mid \mathbf{Y}_i, \mathbf{Z}_i)$ and can therefore implicitly require cross-time or trajectory-level conditional independence. Leaving the entire cluster out avoids this issue for the visit-level target.

7.2 Recommended K -fold procedure

For $k = 1, \dots, K$:

1. Fit all nuisance models on I_k^c .
2. Pre-specify preprocessing, neural-network weights, embedding extraction, basis estimation, variable selection, and tuning parameters using the corresponding analysis set.
3. Predict population-level quantities for subjects in I_k .
4. Compute held-out visit contributions and one score per held-out subject.

Pool the N subject scores and studentise once at the end. Repeated cross-fitting may be used to reduce random fold variability, but the repetition scheme must be included in the bootstrap or variance calculation if scores are averaged across repeated splits.

7.3 What cross-fitting solves and what it does not

Cross-fitting solves the dependence between nuisance fitting and held-out score construction. It does not by itself guarantee small prediction error, correct marginalisation over random effects, or identification of a latent subject effect. Product-rate and cross-term conditions remain necessary.

8 Relevant theory of the original GCM

This section records the original GCM results that the repeated-measures proof should mirror.

8.1 Original univariate statistic

For i.i.d. observations (X_i, Y_i, Z_i) , define

$$\begin{aligned} f_0(z) &= \mathbb{E}(X \mid Z = z), & g_0(z) &= \mathbb{E}(Y \mid Z = z), \\ \varepsilon_i &= X_i - f_0(Z_i), & \xi_i &= Y_i - g_0(Z_i), \end{aligned}$$

and

$$R_i = \{X_i - \hat{f}(Z_i)\}\{Y_i - \hat{g}(Z_i)\}.$$

The statistic is

$$T_n = \frac{n^{-1/2} \sum_i R_i}{[n^{-1} \sum_i R_i^2 - (n^{-1} \sum_i R_i)^2]^{1/2}}.$$

Define

$$\begin{aligned} A_f &= \frac{1}{n} \sum_i \{f_0(Z_i) - \hat{f}(Z_i)\}^2, & A_g &= \frac{1}{n} \sum_i \{g_0(Z_i) - \hat{g}(Z_i)\}^2, \\ B_f &= \frac{1}{n} \sum_i \{f_0(Z_i) - \hat{f}(Z_i)\}^2 \mathbb{E}(\xi_i^2 \mid Z_i), & B_g &= \frac{1}{n} \sum_i \{g_0(Z_i) - \hat{g}(Z_i)\}^2 \mathbb{E}(\varepsilon_i^2 \mid Z_i). \end{aligned}$$

8.2 Theorem 6: null validity

The pointwise result assumes

$$A_f A_g = o_{\mathbb{P}}(n^{-1}), \quad B_f = o_{\mathbb{P}}(1), \quad B_g = o_{\mathbb{P}}(1), \quad 0 < \mathbb{E}(\varepsilon^2 \xi^2) < \infty.$$

Then

$$\sup_t |\mathbb{P}(T_n \leq t) - \Phi(t)| \rightarrow 0.$$

The uniform result additionally requires, over a null class $\mathcal{P}$,

$$\inf_{P \in \mathcal{P}} \mathbb{E}_P(\varepsilon_P^2 \xi_P^2) > 0, \quad \sup_{P \in \mathcal{P}} \mathbb{E}_P|\varepsilon_P \xi_P|^{2+\eta} < \infty,$$

and uniform versions of the prediction-error conditions.

The key numerator expansion is

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_i R_i &= \frac{1}{\sqrt{n}} \sum_i \varepsilon_i \xi_i + \frac{1}{\sqrt{n}} \sum_i \{f_0(Z_i) - \hat{f}(Z_i)\} \{g_0(Z_i) - \hat{g}(Z_i)\} \\ &\quad + \frac{1}{\sqrt{n}} \sum_i \varepsilon_i \{g_0(Z_i) - \hat{g}(Z_i)\} + \frac{1}{\sqrt{n}} \sum_i \xi_i \{f_0(Z_i) - \hat{f}(Z_i)\}. \end{aligned} \quad (18)$$

The four proof components are:

1. an i.i.d. CLT for $n^{-1/2} \sum_i \varepsilon_i \xi_i$;
2. Cauchy–Schwarz for the product-bias term;
3. null conditional mean-zero arguments for the two cross terms, controlled by B_f, B_g ;
4. expansion of $R_i^2 - \varepsilon_i^2 \xi_i^2$, a law of large numbers, and Slutsky’s lemma for denominator consistency.

The same-sample null argument uses

$$\mathbb{E}(\varepsilon_i \mid Y^{(n)}, Z^{(n)}) = 0, \quad \mathbb{E}(\xi_i \mid X^{(n)}, Z^{(n)}) = 0,$$

which follows from i.i.d. sampling and $X \perp\!\!\!\perp Y \mid Z$.

8.3 Theorem 8: alternatives and power

Define

$$\rho_P = \mathbb{E}_P\{\text{Cov}_P(X, Y | Z)\} = \mathbb{E}_P(\varepsilon_P \xi_P), \quad \sigma_P^2 = \text{Var}_P(\varepsilon_P \xi_P).$$

When $\hat{f}$ and $\hat{g}$ are trained on an auxiliary sample independent of the test sample, the centred statistic satisfies

$$\frac{\tau_N^{(n)} - \sqrt{n}\rho_P}{\tau_D^{(n)}} \xrightarrow{d} N(0, 1), \quad \tau_D^{(n)} \rightarrow_{\mathbb{P}} \sigma_P,$$

under analogues of Theorem 6's conditions for $\varepsilon_P \xi_P - \mathbb{E}_P(\varepsilon_P \xi_P)$. Thus the empirical residual covariance estimates ρ_P at the root- n rate.

Auxiliary-sample training is used because, under an alternative, $\mathbb{E}(\varepsilon | Y, Z)$ and $\mathbb{E}(\xi | X, Z)$ need not vanish. Given independent training data and Z , however, the nuisance errors are fixed functions of Z , and

$$\mathbb{E}\{\varepsilon \Delta_g(Z) | Z, \mathcal{T}\} = 0, \quad \mathbb{E}\{\xi \Delta_f(Z) | Z, \mathcal{T}\} = 0.$$

8.4 Theorem 9: multivariate and high-dimensional GCM

For $X \in \mathbb{R}^{d_X}$ and $Y \in \mathbb{R}^{d_Y}$, construct pairwise statistics T_{jk} and define

$$S_n = \max_{j \leq d_X, k \leq d_Y} |T_{jk}|.$$

The reference distribution is the maximum of a centred Gaussian vector with covariance estimated from the empirical correlations of pairwise residual-product vectors.

The high-dimensional theorem requires:

1. sub-exponential or sufficiently high-order moment conditions for every $\varepsilon_j \xi_k$;
2. non-degenerate pairwise variances;
3. a high-dimensional CLT rate such as $C_n^2 \{\log(dn)\}^7/n$ tending to zero sufficiently fast;
4. the strengthened product-rate condition

$$\max_{j,k} A_{f,j} A_{g,k} = o_{\mathbb{P}}\{n^{-1}(\log d)^{-2}\};$$

5. maximal residual and prediction-error conditions controlling all cross terms simultaneously.

The proof combines high-dimensional Gaussian approximation, maximal inequalities for the nuisance terms, sup-norm covariance estimation, and Gaussian anti-concentration.

8.5 Theorem 11: kernel ridge regression

The kernel-ridge section verifies Theorem 6's abstract prediction requirements for a concrete regression method. If f_0, g_0 belong to a reproducing-kernel Hilbert space, conditional variances are bounded, residual-product moments are controlled, RKHS norms are uniformly bounded, and the kernel eigenvalue tails satisfy

$$\limsup_{\lambda \downarrow 0} \sup_{P \in \mathcal{P}} \sum_{j=1}^{\infty} \min(\mu_{P,j}, \lambda) = 0,$$

then the KRR-based GCM is pointwise, and under uniform assumptions uniformly, asymptotically normal.

The proof derives a KRR prediction bound by decomposing error into a variance term and a regularisation-bias term, controls empirical kernel eigenvalues through population eigenvalues, chooses λ to make the prediction MSE $o(n^{-1/2})$, verifies $A_f A_g = o_{\mathbb{P}}(n^{-1})$ and $B_f, B_g = o_{\mathbb{P}}(1)$, and then invokes Theorem 6.

9 Case B: observed, estimated, and latent subject context

9.1 Observed context reduces to Case A

When θ_i is observed independently of the tested outcomes, define $Z_{it}^* = (Z_{it}, \theta_i)$ and apply the A1 cluster-score procedure. Time-invariant genetic information is a typical example: its lack of temporal variation does not prevent it from serving as a conditioning variable. It contributes between-subject information and must remain inside each subject's fold.

When θ_i is estimated, write $\hat{\theta}_i = \theta_i + r_i$. Treating $\hat{\theta}_i$ as if it were observed is justified only if the induced nuisance errors satisfy the same product-rate and cross-term conditions as the other regressions. Estimation from auxiliary information independent of the tested innovations is substantially easier to analyse than estimation from the same X and Y trajectories. The latter can create both non-vanishing measurement error and outcome leakage.

9.2 A different scientific target

The subject-conditional null

$$H_0^{\text{sub}} : X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}, U_i \quad (19)$$

asks whether same-visit deviations remain associated after conditioning on a stable latent subject characteristic U_i . This is not the same question as the population-level null. A shared random effect can violate the population null while (19) remains true.

9.3 Latent-effect identification

If U_i is unobserved, (19) is generally not nonparametrically identified from the observed distribution. Its meaning depends on the chosen latent-variable model, random-effect distribution, and random-effect design. The test is therefore necessarily more model-dependent than the marginal cluster GCM.

9.4 Finite visits and irreducible prediction error

Consider the random-intercept model

$$X_{it} = b_i + e_{it}, \quad b_i \sim N(0, \tau^2), \quad e_{it} \sim N(0, \sigma^2).$$

Even when τ^2 and σ^2 are known, the posterior mean predictor based on m visits has prediction risk

$$\mathbb{E} [\{b_i - \mathbb{E}(b_i \mid X_{i1}, \dots, X_{im})\}^2] = \frac{\tau^2 \sigma^2}{\sigma^2 + m\tau^2}.$$

For fixed m , this is bounded away from zero. Increasing the number of subjects estimates the random-effect distribution more accurately, but does not consistently recover each realised b_i .

9.5 BLUP shrinkage can create residual dependence

Suppose

$$X_{it} = b_i + e_{it}^X, \quad Y_{it} = b_i + e_{it}^Y,$$

with conditional independence given b_i . If separate BLUPs satisfy $\hat{b}_i^X \approx \lambda_X \bar{X}_i$ and $\hat{b}_i^Y \approx \lambda_Y \bar{Y}_i$ with $0 < \lambda_X, \lambda_Y < 1$, then

$$X_{it} - \hat{b}_i^X = (1 - \lambda_X)b_i + \text{noise},$$

and similarly for Y . The residual product contains

$$(1 - \lambda_X)(1 - \lambda_Y)b_i^2,$$

whose expectation is positive. Thus a subject-conditional null can be falsely rejected because the latent effect is only partially removed.

9.6 Why sample splitting is insufficient

Within-subject splitting can prevent a visit from being used both to estimate U_i and to form its own residual. It removes self-influence but does not create additional information about U_i . With a bounded number of visits, the latent-effect prediction error remains non-vanishing and the product of two such errors can induce first-order bias.

If U_i is estimated using other Y visits from the same subject, a same-visit null may not be enough to centre the cross term. One may implicitly require

$$\mathbb{E}(\varepsilon_{it}^X \mid \mathbf{Y}_i, \mathbf{Z}_i, U_i) = 0,$$

which is closer to a trajectory-level conditional-independence assumption.

9.7 Dense-visit regime

A subject-conditional theory becomes more plausible when

$$N \rightarrow \infty, \quad m_{\min} = \min_i m_i \rightarrow \infty,$$

and finite-dimensional random effects are identifiable from the within-subject design. If

$$\widehat{h}_X(U_i) - h_X(U_i) = O_{\mathbb{P}}(m_i^{-1/2}), \quad \widehat{h}_Y(U_i) - h_Y(U_i) = O_{\mathbb{P}}(m_i^{-1/2}),$$

then the corresponding mean squared errors are $O(m_i^{-1})$. A generic root- N GCM product bound requires approximately

$$\frac{\sqrt{N}}{m_{\min}} \rightarrow 0.$$

This dense-longitudinal regime is not representative of studies with only a few visits per subject.

9.8 Recommended interpretation

For sparse repeated-measures data, the subject-conditional GAMM-residual GCM should be presented as a model-based sensitivity analysis rather than the primary general theory. The primary method should target the population-level null and use population predictions plus cluster-robust inference.

10 Case C: bounded and growing observed history

10.1 C1: many subjects with bounded histories

Let H_{it} be a finite vector constructed only from information available before the tested pair, for example $H_{it} = (X_{i,t-1}, Y_{i,t-1})$. For

$$H_0^C : \quad X_{it} \perp\!\!\!\perp Y_{it} \mid Z_{it}, H_{it},$$

define $Z_{it}^* = (Z_{it}, H_{it})$ and apply the subject-cross-fitted procedure. Serial dependence among visits is retained within S_i ; no visit-level independence assumption is introduced. With fixed-dimensional history and independent subjects, this is a reduction to A1 rather than a separate temporal central limit theorem.

The history must nevertheless be scientifically complete for the stated null. If the true process contains $X_{i,t-1} \rightarrow Y_{it}$ but only $Y_{i,t-1}$ is conditioned on, the fitted test targets a different hypothesis. Flexible regression cannot compensate for an omitted state variable.

10.2 C2: growing history in one or a few long sequences

For fixed N and $T \rightarrow \infty$, subject aggregation does not create an increasing number of independent scores. The residual products form a time series. Shi’s GTCM proposal considers the predictive null $Y_{t+1} \perp\!\!\!\perp X_t \mid \{\bar{Y}_t, \bar{Z}_t\}$ and uses regression stability plus weak dependence to control the bias created by training nuisance functions on correlated observations [8]. That argument motivates the A2/C2 branch of the taxonomy, but it is not supplied by the present $\sqrt{N}$ theorem. A temporal implementation would need an explicit long-run variance or martingale/self-normalised reference distribution and a time-respecting fitting rule.

10.3 High-dimensional measurements and learned representations

GAMMs are convenient for smooth low-dimensional context, but the framework is learner-agnostic. A random forest, boosted model, kernel method, neural regression, or other learner may replace a GAMM when its out-of-cluster prediction errors satisfy the stated product-rate and cross-term conditions. For high-dimensional X , including images or dense sensor windows, a representation $\omega(X)$ may be tested instead of the raw object. DNCIT shows how unsupervised, conditionally unsupervised, or transfer-learned representations can be combined with a downstream nonparametric CIT when the representation training cannot create conditional dependence with Y [10]. In clustered data, that requirement is additional to subject-level splitting: any supervised representation selection involving Y must occur using training subjects only.

This composition changes the detectable alternative and may lose information if the embedding is not sufficient. It does not turn the scalar residual covariance into an omnibus test, and it does not yet solve the trajectory-level nulls D or E.

11 Simulation study

11.1 Objectives and design

The simulation study had six linked components: a preliminary numerical check, a formal assessment under the core Case-A1 population null, a broader collection of A1 null settings, an inferential-unit comparison, a frozen extension isolating the information requirements in Cases B and C1, and a supplementary gradient for generated-context accuracy and local power. The original Case-A analyses used three X markers, one continuous Y , subject-level scores, five folds for Design A, and a Gaussian multiplier global test at $\alpha = 0.05$. Each Monte Carlo repetition was stored separately and could be reproduced from its fixed seed. Exact Clopper–Pearson intervals quantify uncertainty in the rejection rates.

Stage 1 used $N = 120$ subjects. Regular regimes had four visits per subject and the sparse-visit regime sampled two to five visits. Observed covariates entered nonlinear mean functions for X and Y . Four regimes distinguished an independent-random-effect population null, correlated random effects that violate the population null while preserving the subject-conditional null, an alternative with correlated visit errors, and a sparse-visit subject-conditional-null stress. The random-effect correlation under population alternatives was 0.7. Each regime had 100 repetitions and 999 multiplier draws.

Stage 2a extended Design A’s core population-null assessment to 1,000 repetitions. The pre-specified criterion required the exact 95% upper confidence limit for the rejection rate to be at most 0.075. Stage 2b pre-specified six additional population-null generators: random slopes, AR(1) within-subject errors with correlation 0.6, unequal visit counts from two to eight, X -informative visit counts, an omitted z -by-time interaction in the X nuisance model, and a combination of these departures from the core setting. Each setting had to yield 1,000 valid repetitions and satisfy the same upper-limit criterion. The nuisance fits give each subject equal total weight, matching the score target under unequal or informative visit counts.

11.2 Why the independent unit matters

A frozen comparison isolated the inferential-unit decision while holding the residuals fixed. Each of 250 independent subjects contributed six visits. Independent subject effects entered the X and Y regressions, and visit innovations followed separate AR(1) processes with correlation 0.55. Linear nuisance regressions were correctly specified and cross-fitted in five subject folds. The subject-score GCM averaged residual products within subject and used $N = 250$ scores; the visit-i.i.d. GCM applied the same residual products as though all 1,500 visits were independent. An oracle benchmark used the known conditional means and subject scores. The two null settings had 1,000 repetitions each, and four alternatives had 500 repetitions each.

Under the base null, rejection rates were 0.060 for the subject-score GCM, 0.057 for the oracle, and 0.216 for the visit-i.i.d. calculation. Under the random-slope null they were 0.055, 0.057, and 0.240, respectively (Figure 2a). The exact 95% interval for the base subject-score row was 0.0461–0.0766. Its upper endpoint exceeded the frozen 0.075 ceiling by 0.0016, so this row did not pass that strict gate; the random-slope row passed. Both subject-score rows closely tracked the oracle, whereas treating visits as independent was severely anti-conservative.

For residual-covariance coefficients 0.05, 0.10, 0.20, and 0.30, subject-score rejection rates were 0.176, 0.456, 0.942, and 1.000; the corresponding oracle rates were 0.182, 0.458, 0.942, and 1.000 (Figure 2b). Because the visit-i.i.d. null calibration was invalid, its power curve is omitted from interpretation. This focused comparison is separate from the frozen Stage-2b random-slope experiment below.

11.3 Rejection rates under the null

Design A rejected in 47 of 1,000 Stage-2a repetitions. The rejection rate was 0.0470 (Monte Carlo standard error 0.0067; exact 95% interval 0.0347–0.0620), and therefore met the pre-specified core-null criterion. Stage 2b completed all 6,000 repetitions, with 1,000 per setting (Table 3). Five settings met the criterion. In the random-slope setting, the method rejected in 65 of 1,000 repetitions, and the exact 95% upper confidence limit was 0.0821, above the threshold of 0.075. Because the criterion was required to hold in every setting, the overall Stage-2b criterion was not met. We report the six settings separately rather than combining them.

Table 3: Design A rejection rates under six population-null settings. The pre-specified criterion required every exact 95% upper confidence limit to be at most 0.075; the random-slope setting did not meet this criterion.

Scenario	Repetitions	Rejections	Rate	Exact 95% interval
Combined stress	1,000	53	0.053	0.0399–0.0688
Informative visits	1,000	48	0.048	0.0356–0.0631
One-sided nuisance misspecification	1,000	52	0.052	0.0391–0.0676
Random slopes	1,000	65	0.065	0.0505–0.0821
Serial AR(1)	1,000	46	0.046	0.0339–0.0609
Unequal visits	1,000	56	0.056	0.0426–0.0721

Design B behaved differently. It rejected 29/100 repetitions in the regime where the population null was false but the subject-conditional null held, and 39/100 in a sparse-visit regime where the subject-conditional null held. The exact 95% intervals were 0.204–0.389 and 0.294–0.493, respectively. These results agree with the shrinkage mechanism surrounding equation (19): preventing self-influence by visit splitting does not consistently recover a latent subject effect from a bounded number of visits. Design B is therefore retained only as a model-based sensitivity calculation.

11.4 Observed context, estimated context, and bounded history

The Case-B/C extension used 1,000 repetitions per scenario, $N = 250$ independent subjects, five subject-level folds, linear nuisance regressions, and a two-sided cluster-score test at $\alpha = 0.05$. Linear regressions were correctly specified by construction so that the experiment isolated the conditioning information rather than regression flexibility. The primary calibration criterion was again an exact 95% upper confidence limit of at most 0.075.

For Case B, a stable Gaussian θ_i affected both X_{it} and Y_{it} . Their innovations were independent under the target null given (Z_{it}, θ_i) . The observed- context analysis rejected in 52/1,000 repetitions (0.052; exact 95% interval 0.0391–0.0676) and passed the criterion. A proxy estimated from 2,000 independent auxiliary measurements gave 54/1,000 rejections (0.054; 0.0408–0.0699). By contrast, a proxy based on four auxiliary measurements and omission of θ_i both rejected in 1,000/1,000 repetitions. The dense-proxy result is a favourable generated-context example, not a general validation of estimated random effects.

For Case C1, X_{it} and Y_{it} depended on $(X_{i,t-1}, Y_{i,t-1})$, and Y_{it} also depended on $X_{i,t-1}$. The current innovations were independent under the target null. Including the complete finite history gave 30/1,000 rejections (0.030; 0.0203–0.0426), which passed the gate and was conservative in this generator. Including only $Y_{i,t-1}$ or omitting history produced 1,000/1,000 rejections. These two rows are not Type-I-error estimates for the reduced nulls: the reduced nulls are false. They show that incomplete history changes the hypothesis being tested. Under the simulated residual- covariance alternatives, the observed-context Case-B test and complete-history Case-C1 test both rejected in all 1,000 repetitions; these deliberately strong alternatives establish implementation sensitivity, not a detection boundary (Table 4).

Table 4: Frozen Case-B and Case-C1 extension. Target-null rows with omitted or incomplete context diagnose a changed conditioning set and must not be read as calibration estimates for the corresponding reduced nulls.

Case	Conditioning analysis	Rejections	Rate	Exact 95% interval
B null	observed θ_i	52/1,000	0.052	0.0391–0.0676
B target null	sparse proxy for θ_i	1,000/1,000	1.000	0.9963–1.0000
B target null	dense proxy for θ_i	54/1,000	0.054	0.0408–0.0699
B target null	θ_i omitted	1,000/1,000	1.000	0.9963–1.0000
B alternative	observed θ_i	1,000/1,000	1.000	0.9963–1.0000
C null	complete (X_{t-1}, Y_{t-1})	30/1,000	0.030	0.0203–0.0426
C target null	only Y_{t-1}	1,000/1,000	1.000	0.9963–1.0000
C target null	history omitted	1,000/1,000	1.000	0.9963–1.0000
C alternative	complete (X_{t-1}, Y_{t-1})	1,000/1,000	1.000	0.9963–1.0000

The supplementary gradient retained the same generators and conditioning sets. For Case B, the proxy was the mean-scale estimate from m auxiliary measurements that were independent of the test-period innovations. With $m = 4, 20, 100, 500$, and 2,000, the target-null rejection rates were 0.999, 0.292, 0.072, 0.059, and 0.058, respectively (Table 5). The exact upper confidence limit at $m = 500$ was 0.07545, missing the frozen 0.075 criterion by 0.00045; at $m = 2,000$ it was 0.07434. This is a generator-specific accuracy curve rather than a universal measurement-count threshold. It shows why an estimated $\hat{\theta}_i$ needs an explicit error-rate condition: an auxiliary estimate does not become valid merely because it has been named as subject context.

Local residual-covariance alternatives used coefficients 0.05, 0.10, 0.20, and 0.30, with 500 repetitions per coefficient. Observed-context Case-B power was 0.344, 0.864, 1.000, and 1.000; complete-history Case-C1 power was 0.602, 0.996, 1.000, and 1.000. The corresponding frozen

null anchors were 0.052 and 0.030. These curves replace the earlier ceiling-only alternative as the useful sensitivity summary. Their levels should not be compared as a ranking of B versus C1 because the two generators contain different within-subject information.

Table 5: Supplementary Case-B proxy-quality and B/C1 local-power gradients. Proxy rows are target-null diagnostics for an independently generated context estimate. Power rows use the complete scientific conditioning set.

Case-B proxy measurements m	4	20	100	500	2,000
Null rejection rate	0.999	0.292	0.072	0.059	0.058
Alternative coefficient δ	0.05	0.10	0.20	0.30	
Case B observed-context power	0.344	0.864	1.000	1.000	
Case C1 complete-history power	0.602	0.996	1.000	1.000	

12 ADNI applications in the many-subject, sparse-visit regime

12.1 Scientific question and data construction

The Alzheimer’s Disease Neuroimaging Initiative (ADNI) is a longitudinal, multisite observational study [12]. The first application examined whether a multivariate pattern of structural brain volumes remained associated with contemporaneous cognitive severity after adjustment for demographic and technical covariates. MRI atrophy patterns have previously been associated with ADAS-Cog13 and subsequent cognitive decline in ADNI [13]. This MRI analysis used an authorised ADNI Study Files snapshot downloaded on 26 July 2026. Participant-level source and derived data remain local and are not redistributed.

Although ADNI is longitudinal, the present asymptotic regime is A1 rather than A2: the number of participants is large and each participant contributes a small number of irregular visits. Subject scores, not the length of one time series, drive inference. Time-invariant observed variables such as APOE genotype enter the conditioning set as observed Case-B context and remain inside the subject-level fold. No analysis conditions on an estimated latent subject effect.

Each ADAS13 assessment date served as an anchor. We selected the nearest QC-eligible FreeSurfer 7 MRI examination within the pre-specified ± 30 -day window. Ties were broken by minimum absolute date gap and then by the earlier MRI date. Before matching, MRI rows had to contain positive, physiologically plausible required volumes and could not carry an explicit overall, ventricular, or hippocampal QC failure. Same-day MRI duplicates were resolved by processing completeness, QC support, field strength, and image identifier without using ADAS13. The X markers were ventricular volume (bilateral lateral and inferior-lateral ventricles plus third and fourth ventricles), bilateral hippocampal volume, and bilateral entorhinal volume. The continuous Y outcome was ADAS13, for which larger values indicate worse cognition. MRI volumes were measured in mm^3 .

The conditioning set Z contained time since baseline, baseline age, sex, education, APOE4, intracranial volume, ADNI protocol, site, MRI field strength, and the MRI–ADAS13 date gap. Concurrent cognitive scales and diagnosis were excluded: they overlap with ADAS13 or may represent downstream disease stage rather than baseline confounding. The estimand is the population-level conditional association between MRI volumes and ADAS13 after adjustment for these demographic and acquisition variables.

Within 30 days, 8,628 complete matched rows from 2,355 subjects were available. To obtain a genuinely repeated-measures application and a common cohort in which the subject-conditional calculation could be evaluated, the analysis cohort required at least five complete visits. It

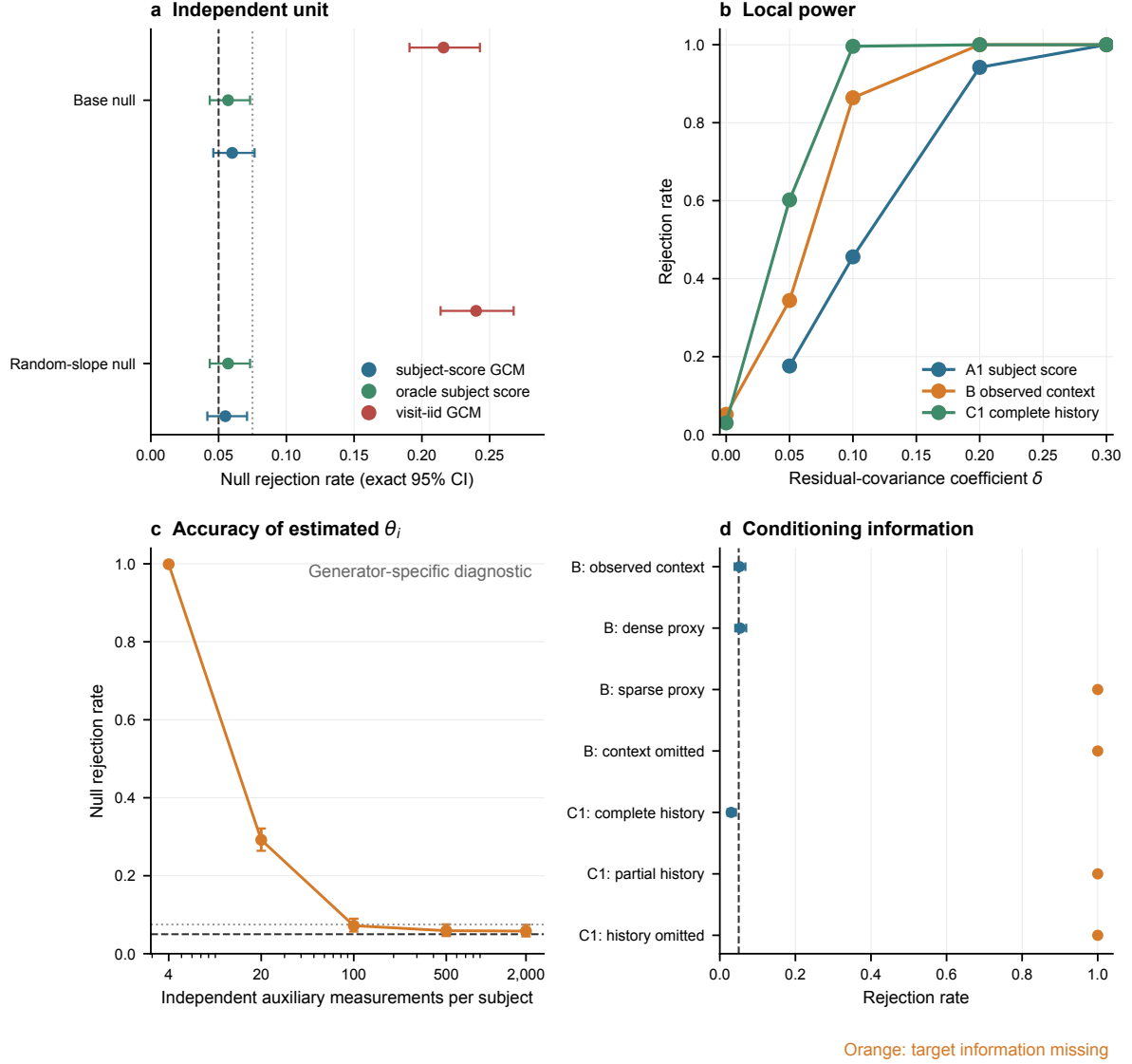


Figure 2: Simulation evidence for the framework boundaries. (a) Subject-score and oracle calculations remain close to the nominal level, while applying the same cross-fitted residual products as independent visits inflates rejection. The dashed and dotted lines mark 0.05 and the frozen 0.075 upper-limit criterion. (b) Local power for the A1 subject-score, observed Case-B, and complete-history C1 constructions; the zero-effect B/C1 points are their frozen null anchors. (c) Case-B target-null rejection as independently measured subject context becomes more accurate; this curve is generator-specific. (d) Information-set diagnostics. Orange rows omit or poorly estimate information in the target conditioning set and therefore do not estimate Type-I error for the stated target null.

contained 4,406 observations from 647 subjects; the median was six visits and the range was five to 13. The median absolute MRI–ADAS13 date gap was one day and its 90th percentile was 19 days. This visit-enriched cohort targets the repeated-measures estimand and represents a selected subset of the full ADNI population.

12.2 Analysis

Continuous nuisance outcomes were fitted with Gaussian identity-link generalised additive models. Smooth terms represented time since baseline, baseline age, and intracranial volume; train-

ing fits included a subject random-effect smooth to account for repeated measurements. Subjects, rather than visits, were assigned to five folds. Held-out predictions excluded the subject random effect, as required for population-level Design A. Scores used $w_{it} = 1/m_i$, giving every subject equal total weight. The global three-marker test used 9,999 Gaussian multiplier draws; marker-wise p -values used a normal reference and Holm correction.

The date window, minimum visit count, covariates, folds, seed, weights, and inferential rules were specified before the first GCM analysis. The bounded-visit simulations did not support Design-B calibration, so real-data inference is restricted to Design A.

12.3 Results

The Design A global test rejected the population-level residual-covariance null ($M = 13.4028$, empirical multiplier $p = 0.0001$). This is the smallest attainable value with 9,999 draws and is not reported as zero. All three marker tests survived Holm correction (Table 6). Larger residual ventricular volume was associated with higher ADAS13, while larger residual hippocampal and entorhinal volumes were associated with lower ADAS13. These directions agree with the medical expectation specified before analysis for atrophy-related cognitive impairment.

Table 6: MRI–ADAS13 Design A results in 647 ADNI participants. Normalised covariances divide the raw equal-subject residual covariance by the two corresponding residual standard deviations. Intervals use the marker-level standard error and a normal approximation; these quantities are descriptive rather than pre-specified effect sizes.

Marker	Normalised covariance	95% interval	T	Holm p
Ventricles	0.307	0.230, 0.384	7.850	4.16×10^{-15}
Hippocampus	−0.466	−0.536, −0.396	−12.979	3.20×10^{-38}
Entorhinal	−0.419	−0.481, −0.358	−13.403	1.75×10^{-40}

Residual means were small relative to their standard deviations: every absolute residual mean was below 4% of its corresponding residual standard deviation. The five folds contained 123–136 subjects and 827–924 visits. Removing any single subject preserved every marker estimate’s direction. Maximum absolute standardised subject scores ranged from 6.46 to 7.47, indicating that influence remains relevant even though no single subject determined a direction. Complete-case filtering, the at-least-five-visit restriction, ADNI participation, and the random-slope simulation result define the population to which this analysis most directly applies.

12.4 What the existing analyses distinguish

The MRI–ADAS13 analysis also provides a descriptive comparison of association magnitudes; predictive biomarker performance requires a separate held-out evaluation. After orienting the three anatomical markers so that larger values represent the adverse direction, the descriptive normalised residual covariances were 0.307 for ventricles, 0.466 for hippocampus, and 0.419 for entorhinal cortex. Thus the hippocampal and entorhinal signals were stronger than the ventricular signal on this residual scale. Because this ordering was examined after the main analysis and denominator uncertainty was not propagated into formal between-marker comparisons, it is treated as hypothesis-generating rather than as a ranking of clinical biomarkers.

An earlier cluster-GCM analysis aligned the same three MRI regions to AV45 amyloid PET in 672 observations from 467 participants. It provides a sharper demonstration of the incremental-information estimand. When concurrent cognition and baseline covariates were in the reference set, the global test did not reject ($M = 1.7663$, $p = 0.2124$), and no marker survived Holm correction (Holm $p = 0.8185$, 0.2320, and 0.8185 for ventricles, hippocampus,

and entorhinal cortex). In a distinct sensitivity estimand that omitted concurrent cognition, the global test rejected ($M = 6.3355$, empirical $p = 0.0001$), and all three markers survived Holm correction. Their direction-oriented normalised residual covariances were 0.134, 0.277, and 0.203, respectively. The two results cannot be pooled: they show that whether MRI is judged to add information depends on what information is already represented by Z .

Table 7: Extent to which the completed ADNI experiments address the core incremental-information objective. Each conclusion is relative to its named candidate, target, and reference set.

Completed analysis	Relation to the core objective	Result and interpretation
MRI–AV45, cognition-conditioned	Direct test of whether regional MRI (X) contains continuous amyloid-PET information (Y) beyond cognition and baseline/technical context (Z).	Global $p = 0.2124$; no marker survived Holm correction. No detectable incremental MRI–amyloid information remained beyond the measured cognitive context.
MRI–AV45, baseline-covariate sensitivity	Comparator with cognition removed from Z , showing how the answer changes when behavioural context is unavailable.	Global empirical $p = 0.0001$; all markers survived Holm correction with coherent directions. MRI carried total residual amyloid signal under this smaller reference set.
MRI–ADAS13	Regional MRI (X) and concurrent ADAS13 (Y) after demographic and technical adjustment; a supporting application rather than the pathology-beyond-cognition question.	Global empirical $p = 0.0001$; all markers survived Holm correction. Supports coherent contemporaneous MRI–cognition signal and method feasibility; pathology-specific and prospective value are addressed by the other designs.
Plasma–Centiloid, cognition/genetics-conditioned	Direct test of whether four accessible plasma markers contain amyloid-PET information beyond cognition, demographics, APOE, and acquisition context.	Global empirical $p = 0.0001$. Unique-marker tests retained p-tau217 and A β 42/40 but not NfL or GFAP; adding plasma reduced held-out MSE by 29.7%. The same marker distinction held in 20/20 repeated subject-fold assignments. This is the strongest completed match to the core objective.
Plasma–future ADAS13, history-conditioned	Prospective test of whether plasma markers contain prognostic information beyond current cognition, prior cognition, demographics, APOE, and timing.	Global empirical $p = 0.0001$. Only p-tau217 retained a unique signal after adjustment for the other markers. Its held-out MSE reduction was 15.2% in the 30-day cohort and 17.3% in the 90-day sensitivity cohort; repeated-fold direction and prediction gains were stable, with adjusted detection in 19/20 splits.

The completed MRI–AV45 experiment addresses the core objective for one candidate modality, one pathology target, and one measured context set. Its primary answer is that no additional MRI information about amyloid was detected after cognition was included. This separates a plausible marginal MRI–amyloid signal from incremental information in the analysed cohort. Concurrent cognition may represent shared disease information, mediation, overadjustment, or collider distortion, so the reference-set comparison is itself scientifically informative.

The applications support three levels of interpretation. The global test asks whether the marker panel contributes any residual information. Marker-wise tests identify which pre-specified measurements contribute detectable residual signal. A stronger claim that one marker

provides unique or practically useful information requires conditioning on the other markers and demonstrating out-of-subject predictive improvement. The MRI analyses provide the first two levels; the plasma analyses add unique-marker tests and held-out prediction on common cohorts (Table 7).

12.5 Plasma information about PET amyloid beyond cognition and genetics

We next implemented the pre-specified primary pathology-information combination. Amyloid PET acquisitions were the anchors, with continuous Centiloids as Y . The accessible plasma block X comprised p-tau217, A β 42/40, NfL, and GFAP. The reference set Z comprised concurrent MOCA, MMSE, and CDR-SB; age, sex, education, APOE ϵ 4 dosage, PET tracer, visit time, and exact plasma and cognitive date gaps. Plasma and cognition were deterministically matched within ± 90 days. Negative assay missing-value codes were removed before complete-case construction. The resulting common cohort contained 758 PET-anchored observations from 664 participants; 94 participants had two observations. The median absolute plasma–PET gap was 7 days, its 90th percentile was 42 days, and the maximum was 90 days.

Because most participants contributed one observation, the Design A nuisance models omitted the computationally redundant subject random-effect smooth, but retained equal-subject training weights, participant-level folds, participant-level score aggregation, and $\sqrt{N}$ scaling. This does not change the population-level estimand. The global four-marker test rejected the conditional residual-covariance null ($M = 9.3130$, multiplier $p = 0.0001$). On the marker-wise analysis, p-tau217, A β 42/40, and GFAP survived Holm correction, whereas NfL did not (Holm $p = 0.7557$). Their residual-covariance signs were medically coherent: higher p-tau217 and GFAP and lower A β 42/40 accompanied higher residual Centiloids.

The stronger candidate-discrimination analysis placed the other three plasma markers in the reference set for each marker in turn. P-tau217 retained a normalised residual covariance of 0.505 (Holm $p = 9.19 \times 10^{-16}$), and A β 42/40 retained -0.264 (Holm $p = 8.25 \times 10^{-12}$). By contrast, NfL (-0.098 , Holm $p = 0.1571$) and GFAP (0.008 , Holm $p = 0.8559$) did not retain detectable unique amyloid information after the other plasma markers were included. Thus the result distinguishes an amyloid-focused pair from two markers whose total signal was shared with that pair. This marker-level separation is consistent with external evidence that plasma A β 42/40 and p-tau217 play distinguishable roles in amyloid-related trial selection and disease monitoring [15]; that agreement is a biological plausibility check rather than part of the selection rule (Table 8).

Held-out prediction supplied a separate practical check. Adding the four plasma markers to the same reference model reduced participant-level standardised-outcome RMSE from 0.794 to 0.666, equivalent to a 29.7% mean-squared-error reduction. A participant bootstrap gave a positive mean MSE improvement interval of 0.126–0.250 and one-sided $p = 0.0001$. The global test, unique-marker pattern, and held-out gain therefore agree that plasma is the completed ADNI combination that best instantiates the paper’s core objective.

Twenty additional subject-fold assignments held the cohort, variables, and formulas fixed. P-tau217 remained positive and Holm-significant in 20/20 assignments, with normalised covariances from 0.456 to 0.497; A β 42/40 remained negative and significant in 20/20, from -0.276 to -0.244 . NfL and GFAP were detected in 0/20 assignments. The plasma-block MSE reduction was positive in every assignment, with a median of 27.9% and a range of 25.4%–29.3%. These repeated folds quantify algorithmic stability within the selected cohort.

A direct cross-modality check used the 570 participants (641 observations) with both plasma and a QC-eligible structural MRI within 90 days of PET. After moving the MRI block into Z , the plasma block remained detectable (global $p = 0.0001$); after moving the plasma block into Z , the MRI block did not ($p = 0.2378$). On the same folds and cohort, plasma reduced held-out MSE by 28.8%, MRI by 0.3%, and both together by 29.2%. This common-cohort result, rather than unequal-sample p -value ranking, is the basis for selecting plasma–Centiloid as the current

primary real-data experiment. Figure 3 places this common-cohort comparison next to the unique-marker and prospective results.

Table 8: Unique-marker plasma–amyloid results after conditioning on cognition, genetics, acquisition context, and the other three plasma markers.

Marker	Normalised covariance	Holm p	Incremental-information conclusion
p-tau217	0.505	9.19×10^{-16}	detected
A β 42/40	−0.264	8.25×10^{-12}	detected
NfL	−0.098	0.1571	not detected
GFAP	0.008	0.8559	not detected

12.6 Prospective plasma information about future cognition given observed history

We next used the bounded observed-history branch C1 to address prognosis. Plasma assay dates served as candidate index dates. The marker block X contained log-transformed p-tau217, A β 42/40, NfL, and GFAP. The outcome Y was a future ADAS-Cog13 assessment between 270 and 1,278 days after the index. When several assessments were available, we selected the one closest to 730 days. The reference set Z contained age, sex, education, APOE ϵ 4 dosage, concurrent ADAS-Cog13, and the current and future date gaps. The observed history H_{it} contained the most recent ADAS-Cog13 score from 180–1,080 days before the index and its date gap. APOE was observed stable Case-B context rather than an estimated latent effect.

We retained the earliest eligible index date for each participant. The primary ± 30 -day current-assessment window produced 199 independent participant rows. The median absolute current gap was zero days. The median prior-assessment gap was 384 days, and the median future-assessment gap was 526 days. The future-gap interquartile range was 373–740 days. A pre-specified ± 90 -day sensitivity window produced 203 participants. Subject-level five-fold cross-fitting was retained, although each selected participant contributed one index row.

The primary four-marker maximum test rejected the history-conditioned residual-covariance null ($M = 4.2330$, multiplier $p = 0.0001$). Each unique-marker test also conditioned on the other three plasma measurements. P-tau217 retained a normalised residual covariance of 0.321 (Holm $p = 0.00123$). A β 42/40, NfL, and GFAP did not retain detectable unique information (Table 9). Deleting any one participant left the p-tau217 unique-marker p -value between 8.01×10^{-5} and 5.59×10^{-4} . The ± 90 -day sensitivity analysis gave the same marker distinction, with a p-tau217 normalised covariance of 0.285 and Holm $p = 0.000380$. Independent multicohort work has likewise found that plasma p-tau217 is associated with future cognitive decline and can complement tau PET [16]. The present analysis isolates its incremental information in the specified ADNI cohort and conditioning set.

Across 20 additional subject-fold assignments in the primary cohort, the p-tau217 unique covariance remained positive, with a median of 0.297 and a range of 0.230–0.351. A conservative four-marker Bonferroni adjustment remained below 0.05 in 19/20 assignments; the largest adjusted value was 0.098. The direction is therefore stable, while the binary significance decision shows mild split sensitivity in this smaller cohort.

Held-out prediction provided a complementary practical comparison. Adding p-tau217 to the history-conditioned reference reduced RMSE from 5.851 to 5.387, a 15.2% MSE reduction. The participant bootstrap interval for the mean MSE reduction was −0.496 to 11.006, with a one-sided $p = 0.0375$. The interval crosses zero, so the primary prediction gain is suggestive under a two-sided 95% criterion. In the 90-day sensitivity cohort, RMSE fell from 5.886 to 5.351. The corresponding 17.3% MSE reduction had a bootstrap interval of 0.334–12.231 and one-sided $p = 0.0186$. By contrast, adding prior cognition to the current-cognition reference

Table 9: Prospective unique-marker results for future ADAS-Cog13. Every test conditions on current cognition, prior cognition, demographics, APOE, timing, and the other plasma markers.

Marker	Primary covariance	Primary Holm p	90-day Holm p	Unique information
p-tau217	0.321	0.00123	0.000380	detected
A β 42/40	0.096	0.3988	0.2078	not detected
NfL	0.057	0.7362	0.7131	not detected
GFAP	-0.054	0.7362	0.8215	not detected

reduced MSE by only 1.9% in the primary cohort. The history remains part of the conditioning set because it defines the scientific null, not because it must improve prediction strongly. Across the 20 repeated fold assignments, the p-tau217 MSE reduction was positive every time, with a median of 21.0% and a range of 14.6%–26.6%.

This prospective result distinguishes candidate markers instead of reporting only association directions. P-tau217 was the only marker supported by the unique-information test, the leave-one-participant-out check, and held-out prediction. The analysis belongs to the many-subject C1 branch, which reduces to A1 after adding fixed observed history; the single-series temporal theory discussed by Shi instead motivates the growing-history C2 branch.

12.7 Additional ADNI conditional pathology-information targets

The completed plasma-amyloid experiment retains the scientific structure of the original IWSM example: does an accessible candidate measurement X contain information about established disease pathology Y beyond clinical or behavioural context Z ? The remaining MRI block contains hippocampal, entorhinal, and ventricular measures. A second pathology target is temporal-meta-ROI tau PET SUVR, for which amyloid burden is added to Z so that the target is tau information beyond amyloid stage. The reference set also contains contemporaneous cognition, age, sex, education, site/phase, visit time, assay and imaging versions, and exact date gaps. APOE ϵ 4 dosage and a frozen externally weighted non-APOE polygenic score form a time-invariant genetic background block G_i . They enter Z for plasma and MRI tests. In a reciprocal analysis, G_i is instead the candidate X and is not repeated as though each visit supplied a new genotype observation.

PET acquisition dates anchor the primary alignment. Plasma, MRI, and cognition are matched using continuous dates inside a frozen ± 90 -day window; ± 180 days is a sensitivity analysis. All visits for one participant remain in one fold, and repeated residual products are aggregated to one equal-subject score. The APOE-adjusted cohort is analysed without requiring a non-APOE score for everyone; the polygenic extension has its own documented complete-case cohort.

Candidate discrimination was pre-specified at five levels:

1. a global test across all frozen candidates;
2. multiplicity-adjusted plasma, MRI, and genetic block tests;
3. unique-block tests that move the other candidate blocks into Z ;
4. Holm-adjusted marker-wise and within-block unique-marker tests; and
5. participant-held-out changes in RMSE and MAE for continuous PET, with AUC and Brier-score changes for a secondary frozen PET-positivity threshold.

Candidate ranking is based on a common comparison cohort, cross-fitted scale-normalised residual covariances, participant-bootstrap intervals for their pairwise differences, unique-

information tests, and paired prediction-loss changes. Maximal cohorts for individual blocks are sensitivity analyses rather than a basis for ranking unequal-sample p-values.

The completed amyloid result supplies the first candidate-by-target information signature rather than a list of signs: p-tau217 and A β 42/40 retained unique amyloid information, whereas NfL and GFAP did not. For the remaining target, plasma p-tau217 is expected to carry tau information; NfL and structural MRI should be more closely related to neurodegeneration and subsequent cognition; and genetic risk may attenuate once measured pathology enters Z [14, 17]. These frozen qualitative expectations serve as external plausibility checks while leaving discordant results reportable. An ADNI4 remote extension can treat Storyteller speech features or ECog-12 as accessible X , standard cognitive measures and genetics as Z , and blood or PET pathology as Y . It is promoted only if continuous-date linkage and pathology overlap are adequate [18].

Together, the prospective cognitive target and the pathology-information target preserve the link between the IWSM example and future wearable applications. Tau PET and a frozen non-APOE polygenic score are the next conditional pathology-information targets.

13 Discussion

The central contribution is a separation principle: the target null, the available conditioning information, and the independent asymptotic unit must be chosen before the nuisance learner. The five cases make this principle operational. A1 uses independent subject scores; A2 and C2 derive information from time; A3 needs two-rate asymptotics; B can contain a generated-variable or latent-variable identification problem; C1 can reduce to A1; and D/E concern stochastic paths. Calling all of these settings “mixed GCM” would obscure the change in target and reference distribution.

Within A1, subject aggregation aligns the score, variance estimate, and cross-fitting unit. The direct method comparison makes the consequence visible: using the same cross-fitted residuals but treating visits as independent produced null rejection rates above 0.20, whereas subject scores closely tracked the oracle. The wider simulation programme also defines the current boundary. The core null and five of six Stage-2b settings met the frozen criterion, while the original random-slope stress setting did not. Observed Case-B context and complete C1 history calibrated well; a dense auxiliary proxy succeeded in its favourable generator, but a sparse proxy and incomplete history did not recover the target null. The new proxy gradient sharpens that statement: calibration improved continuously with auxiliary accuracy, and even 500 measurements narrowly missed the strict frozen gate in this generator. The local-power curves show useful sensitivity before the ceiling effects of the original strong alternatives.

The ADNI analyses show why the conditioning set is scientifically consequential. Structural MRI retained coherent contemporaneous information about ADAS-Cog13, but did not retain detectable amyloid-PET information once concurrent cognition entered the reference set. In the primary pathology experiment, p-tau217 and A β 42/40 retained unique information about Centiloid, while NfL and GFAP did not. On a common cohort, plasma reduced held-out MSE by 28.8%, compared with 0.3% for structural MRI. In the prospective C1 analysis, p-tau217 was the only plasma marker with unique information about future ADAS-Cog13 and reduced held-out MSE by 15.2%–17.3%. Repeated subject folds preserved the plasma–amyloid two-marker distinction and positive prediction gain in every assignment. The prospective p-tau217 direction and prediction gain were also stable, although one of 20 assignments did not cross the adjusted 0.05 threshold. The contribution is therefore a candidate-information signature relative to a named reference set, rather than recovery of plausible signs alone.

GTCM and DNCIT identify two natural extensions. GTCM supplies a route to temporally dependent residual scores for growing-history C2; DNCIT supplies conditions for composing a learned representation of high-dimensional X with a downstream CIT. A future hybrid must

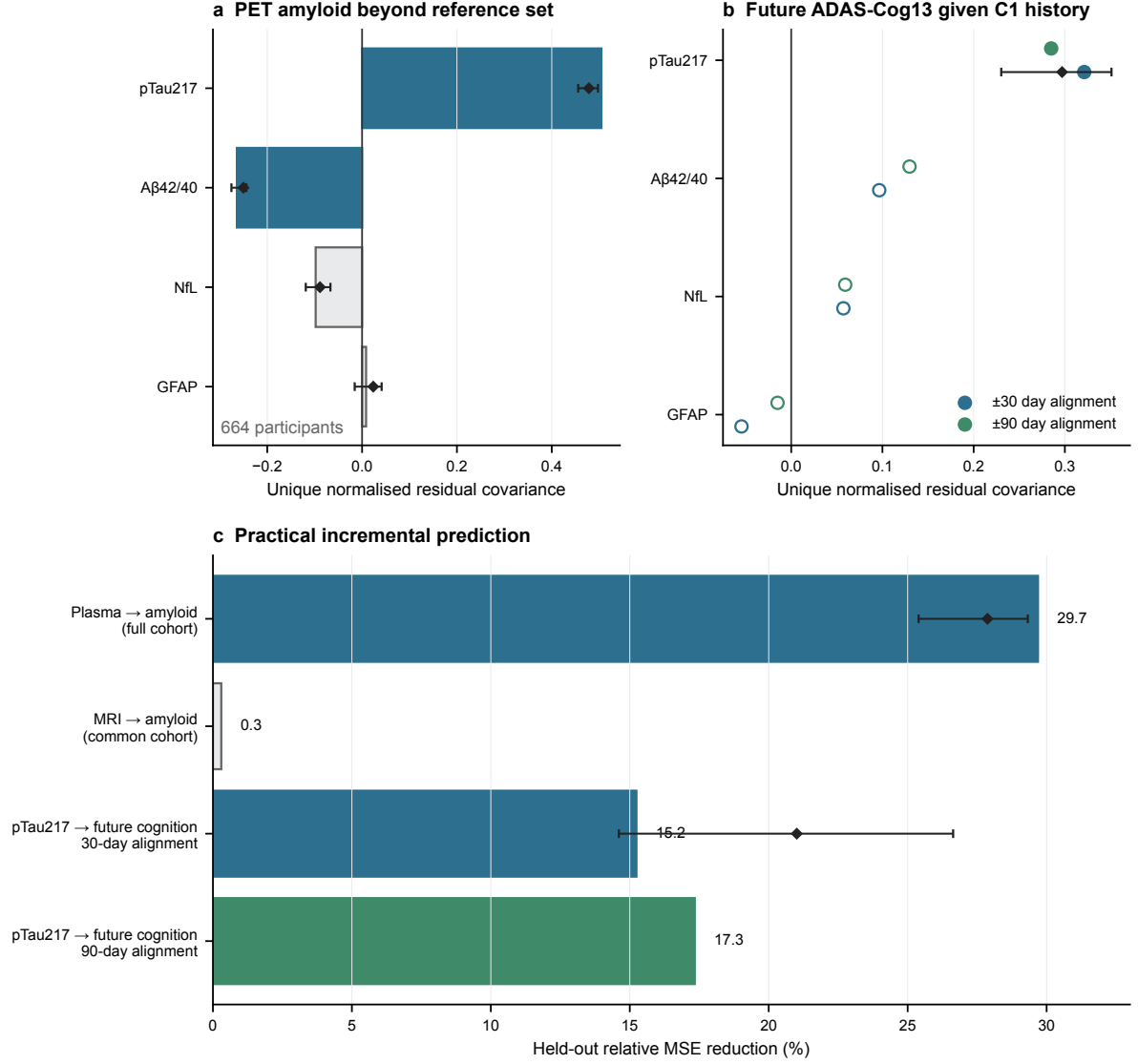


Figure 3: ADNI candidate-information signatures rather than sign recovery alone. (a) Unique plasma-marker residual covariances with PET amyloid after conditioning on cognition, demographics, APOE, acquisition context, and the other plasma markers; blue bars survive Holm correction. Black diamonds and whiskers show the median and range across 20 additional subject-fold assignments. (b) Unique plasma information about future ADAS-Cog13 after current and prior cognition (C1); filled points survive Holm correction. (c) Participant-held-out prediction gains. Plasma added substantially more predictive information than structural MRI, and p-tau217 improved future-cognition prediction under both alignment windows. Repeated-fold diamonds and whiskers are shown for the full plasma cohort and primary future-cognition analysis.

satisfy both representation-training and non-i.i.d. calibration conditions. The motivating private wearable application has the same scientific form—whether device signals contain outcome information beyond recorded activity—but no private wearable data or result enters this article.

Three limitations remain central. The A1 result is currently a high-level theorem and proof roadmap rather than a submission-complete technical proof. The scalar residual covariance is not omnibus against conditional-dependence alternatives with zero conditional covariance. Finally, the ADNI findings are observational, complete-case, and exploratory; they establish

neither causality nor clinical biomarker qualification. These limits determine the next theoretical and empirical steps rather than changing the five-case taxonomy.

14 Reproducibility, data availability, and governance

The analysis code, pre-specified settings, separately stored Monte Carlo repetitions, summary tables, and dated validation reports are organised in the accompanying EGCM repository [7] and at <https://github.com/Aokowww/EGCM>. The MRI-ADAS13 Design A analysis recorded SHA-256 hashes for the analysis table, settings, complete results, and summary tables. The analytic input contained 4,406 rows from 647 subjects; its SHA-256 hash begins `bb426c9b0df2`, with the complete value retained in the dated experiment record. ADNI source and participant-level derived files are excluded from version control and cannot be redistributed. Reproduction of the application requires separately approved ADNI access and the documented source-table versions. The plasma-amyloid analysis used the 12 August 2026 UPENN plasma and UC Berkeley amyloid-PET Study Files snapshots. Restricted rows were processed locally; only aggregate, disclosure-checked cohort, test, unique-marker, and prediction summaries are retained in the public repository. Its analytic table contained 758 rows from 664 participants; the SHA-256 begins `d218c64b6c0d`. Remaining analyses must follow the same restricted-data boundary. The prospective analysis used a local ADNIMERGE2 archive obtained on 23 August 2026 and wrote no participant-level derived table. Its code and frozen configuration hashes begin `5ff00256d72b` and `e8d930856e5e`. Public cohort, marker, and prediction files contain aggregate rows only. The repeated-fold ADNI extensions retained only split-level aggregate statistics. For the plasma-amyloid analysis, the configuration, runner, and public summary hashes begin `1327be144ff8`, `68b842d657e4`, and `0c00c0e32d6d`. For prospective C1, the corresponding hashes begin `70bca1df4fc2`, `b4e2be9b7035`, and `28c5917892c5`. No fold membership, residual, row-level prediction, identifier, or date is released.

The Case-B/C extension was frozen before its 1,000-repetition run. Its configuration, simulation script, dated plan, aggregate summary, replicate-level results, and validation report are retained separately. The SHA-256 hashes of the configuration and aggregate summary begin `ae66ba192f65` and `beb66ef1df74`, respectively. The experiment contains synthetic data only. No wearable-study row or other private application data were used.

The supplementary B/C1 gradient comprised 9,000 synthetic data sets. The SHA-256 hashes of its configuration, runner, aggregate summary, and replicate-level results begin `0e8d54662bcf`, `cb83c321b8e7`, `7219a60e54cf`, and `fc134d143ad0`. Only the aggregate summary is included in the public artifact.

The inferential-unit comparison was frozen before its 4,000 simulated data sets were run. The SHA-256 hashes of its configuration, simulation script, and full aggregate summary begin `69a947f1cc25`, `1ce8fa18bbd2`, and `218e3c4b6098`, respectively. Replicate-level results remain outside the public release; the disclosure-safe aggregate summary and validation record are public.

ADNI source data are available to approved investigators through the Image and Data Archive, subject to the ADNI Data Use Agreement and publication policy. The present work is a secondary analysis of de-identified data and involved no new participant contact or data collection. ADNI study protocols were approved by the participating sites and written informed consent was obtained under those protocols [12].

15 Limitations and next directions

The first priority is a submission-complete proof of the A1 result, including the oracle cluster central limit theorem, cross-fitted remainder bounds, score-variance consistency, and fixed and

local alternatives. Any response to the original Stage-2b random-slope failure must be evaluated under a newly frozen method version rather than tuned against that row.

A2 and A3 require separate theory. A2 needs a temporal score process, weak-dependence or martingale conditions, and long-run variance or self-normalised calibration. A3 must state joint growth conditions for N , T_i , and the largest cluster contribution. In Case B, sufficient rates are needed when $\hat{\theta}_i$ is estimated from independent auxiliary data, together with additional leakage terms when it is learned from the same trajectory. Sparse BLUPs remain a negative control unless a model-specific identification theorem is available.

For C2, future simulations should vary history length, mixing strength, proxy accuracy, and local alternative size, followed by a growing-history theorem comparable in purpose to the GTCM stability argument. Learner-specific corollaries for GAMMs, kernel ridge regression, and other regressions can be stated through common prediction-rate conditions. A high-dimensional extension must additionally combine DNCIT-style representation separation with subject-level splitting and non-i.i.d. calibration.

Cases D and E require a path-space estimand and a trajectory-level score before calibration can be addressed. Empirically, tau-PET, genetic, and independent ADNI snapshot analyses should be frozen separately from the completed same-time and future-cognition experiments. Repeated folds quantify algorithmic sensitivity but do not replace temporal or external-cohort validation. Only aggregate, disclosure-checked outputs may enter public artifacts.

16 Scope and evidence summary

Case	Scientific target	Theoretical status	Empirical evidence	Interpretation
A1	state-level, many subjects	developed proof programme under cluster rates	Stage 2a; six stress settings; ADNI	primary current scope
A2	state/predictive, long sequence	requires temporal theory	Shi/GTCM comparison only	not implemented here
A3	state-level, both dimensions grow	requires double asymptotics	none	future theory
B	state-level given θ_i	observed case reduces to A; latent case conditional	observed context, proxy-quality gradient, and local power	heuristic if estimated
C1/C2	state-level given history	bounded-history C1 reduces to A1; growing-history C2 is temporal	C1 calibration/local power plus prospective ADNI and repeated folds	regime-dependent
D/E	trajectory-level paths	no theorem here	none	future work

Acknowledgements

Data used in preparation of this article were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database. As such, the investigators within ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at https://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf.

Data collection and sharing for ADNI are funded by the National Institute on Aging (National Institutes of Health Grant U19AG024904). The grantee organisation is the Northern California Institute for Research and Education. In the past, ADNI also received funding from the National Institute of Biomedical Imaging and Bioengineering, the Canadian Institutes of

Health Research, and private-sector contributions through the Foundation for the National Institutes of Health, including AbbVie; Alzheimer’s Association; Alzheimer’s Drug Discovery Foundation; Araclon Biotech; BioClinica, Inc.; Biogen; Bristol-Myers Squibb Company; CereSpir, Inc.; Cogstate; Eisai Inc.; Elan Pharmaceuticals, Inc.; Eli Lilly and Company; EuroImmun; F. Hoffmann-La Roche Ltd and its affiliated company Genentech, Inc.; Fujirebio; GE Healthcare; IXICO Ltd.; Janssen Alzheimer Immunotherapy Research & Development, LLC; Johnson & Johnson Pharmaceutical Research & Development LLC; Lumosity; Lundbeck; Merck & Co., Inc.; Meso Scale Diagnostics, LLC; NeuroRx Research; Neurotrack Technologies; Novartis Pharmaceuticals Corporation; Pfizer Inc.; Piramal Imaging; Servier; Takeda Pharmaceutical Company; and Transition Therapeutics.

References

- [1] R. D. Shah and J. Peters. The hardness of conditional independence testing and the generalised covariance measure. *The Annals of Statistics*, 48(3):1514–1538, 2020.
- [2] K. Fukumizu, A. Gretton, X. Sun, and B. Schölkopf. Kernel measures of conditional dependence. In *Advances in Neural Information Processing Systems 20*, pages 489–496, 2007.
- [3] K. Zhang, J. Peters, D. Janzing, and B. Schölkopf. Kernel-based conditional independence test and application in causal discovery. In *Proceedings of the 27th Conference on Uncertainty in Artificial Intelligence*, pages 804–813, 2011.
- [4] C. Scheidegger, J. Hörrmann, and P. Bühlmann. The weighted generalised covariance measure. *Journal of Machine Learning Research*, 23(273):1–68, 2022.
- [5] A. Bellot and M. van der Schaar. Conditional independence testing using generative adversarial networks. In *Advances in Neural Information Processing Systems 32*, 2019.
- [6] K. Chwialkowski and A. Gretton. A kernel independence test for random processes. In *Proceedings of the 31st International Conference on Machine Learning*, volume 32 of *Proceedings of Machine Learning Research*, pages 1422–1430, 2014.
- [7] Q. Huo, M. Simnacher, X. Xu, and S. Greven. Extending the generalised covariance measure for conditional independence testing in repeated-measures data. In *Proceedings of the 40th International Workshop on Statistical Modelling*, 2026.
- [8] J. Shi. Conditional independence testing in time series: a primer. Statistical Science Seminar, University College London, 5 February 2025.
- [9] M. Wieck-Sosa, M. F. C. Haddad, and A. Ramdas. Conditional independence testing with a single realization of a multivariate nonstationary nonlinear time series. *arXiv preprint arXiv:2504.21647*, 2025.
- [10] M. Simnacher, X. Xu, H. Park, C. Lippert, and S. Greven. Deep nonparametric conditional independence tests for images. *arXiv preprint arXiv:2411.06140*, 2024.
- [11] S. N. Wood. *Generalized Additive Models: An Introduction with R*. Chapman and Hall/CRC, second edition, 2017.
- [12] M. W. Weiner et al. The Alzheimer’s Disease Neuroimaging Initiative: progress report and future plans. *Alzheimer’s & Dementia*, 6(3):202–211.e7, 2010.

- [13] S. L. Risacher, W. H. Anderson, A. Charil, P. F. Castelluccio, S. Shcherbinin, A. J. Saykin, A. J. Schwarz, and the Alzheimer’s Disease Neuroimaging Initiative. Alzheimer disease brain atrophy subtypes are associated with cognition and rate of decline. *Neurology*, 89(21):2176–2186, 2017.
- [14] S. Janelidze et al. Plasma phosphorylated tau 217 and A β 42/40 to predict early brain A β accumulation in people without cognitive impairment. *JAMA Neurology*, 81(9):947–958, 2024.
- [15] N. J. Ashton et al. Differential roles of A β 42/40, p-tau231 and p-tau217 for Alzheimer’s trial selection and disease monitoring. *Nature Medicine*, 28:2555–2562, 2022.
- [16] R. Ossenkoppele et al. Plasma p-tau217 and tau-PET predict future cognitive decline among cognitively unimpaired individuals: implications for clinical trials. *Nature Aging*, 5:883–896, 2025.
- [17] H. Moon, X. Chen, and the Alzheimer’s Disease Neuroimaging Initiative. Plasma p-tau217 predicting brain-wide tau accumulation in preclinical AD. *The Journal of Prevention of Alzheimer’s Disease*, 12(7):100252, 2025.
- [18] C. Skirrow et al. Storyteller in ADNI4: application of an early Alzheimer’s disease screening tool using brief, remote, and speech-based testing. *Alzheimer’s & Dementia*, 20(10):7273–7290, 2024.